

BT101

Introductory Biology

(Lecture 9)

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Introduction



Evolution of life:

- ❑ Origin of Life;
- ❑ Darwin's concepts of evolution;
- ❑ Biodiversity.

Cell, the structural and functional unit of life:

- ❑ Three domains of life;
- ❑ Cell types, cell organelles and structure;
- ❑ Basic biomolecules of cell.

Three domains of life

The tree of life consists of three domains: Bacteria, Archaea and Eukarya

- ❑ All living organisms consist of elementary units called cells.
- ❑ All cells are believed to have evolved from a common progenitor because the structures and molecules in all cells have so many similarities.
- ❑ In recent years, detailed analysis of DNA sequences from a variety of Prokaryotic organisms revealed that they are of **two distinct** types: so-called “True” bacteria or **Eubacteria** and **Archaea**.
- ❑ Based on the assumption that the organisms with more similar genes evolved from a common progenitor more recently than those with more dissimilar genes. The researchers developed an Evolutionary Lineage Tree as shown in Figure 1.
- ❑ According to this tree, the **Archaea** and the **Eukaryotes** diverged from the **Eubacteria** and later they also diverged from each other.

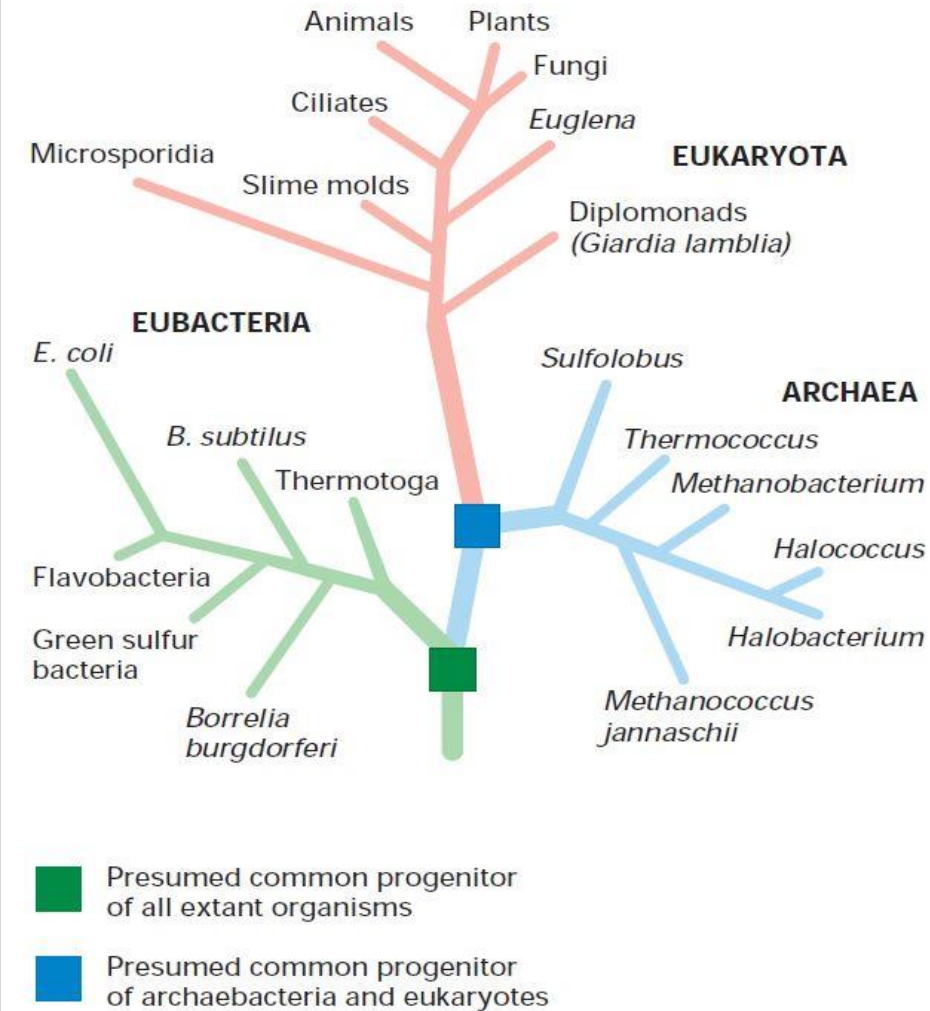


Fig. 1. All organisms from simple bacteria to complex mammals probably evolved from a common, single-celled progenitor. *Credit: Molecular Cell Biology - Lodish

The Cell



- The cell is the basic unit of life. Microorganisms such as Bacteria, Yeast and Amoeba exist as single cells. By contrast, an adult human is made up of around ~37.2 trillion cells (1 trillion = 10^{12}), which are mostly organized into tissues and organs.
- The first person to observe and record cells was Robert Hooke (1635–1703)
- Antonie Van Leeuwenhoek is known for his pioneering work in Microscopy.



Important characteristics of a cell

1. All cells store their hereditary information in the same chemical code: DNA.
2. All cells replicate their hereditary information by DNA template polymerization: Replication.
3. All cells transcribe portions of their hereditary information into an intermediary form: RNA.
4. All cells translate RNA into protein: Translation
5. All cells use proteins as catalysts.
6. Each protein is encoded by a specific gene (segment of DNA).
7. All cells are enclosed in a plasma membrane allowing the nutrients and waste materials exchange.

Introduction



Table 1.1 Prokaryotic and Eukaryotic Cells

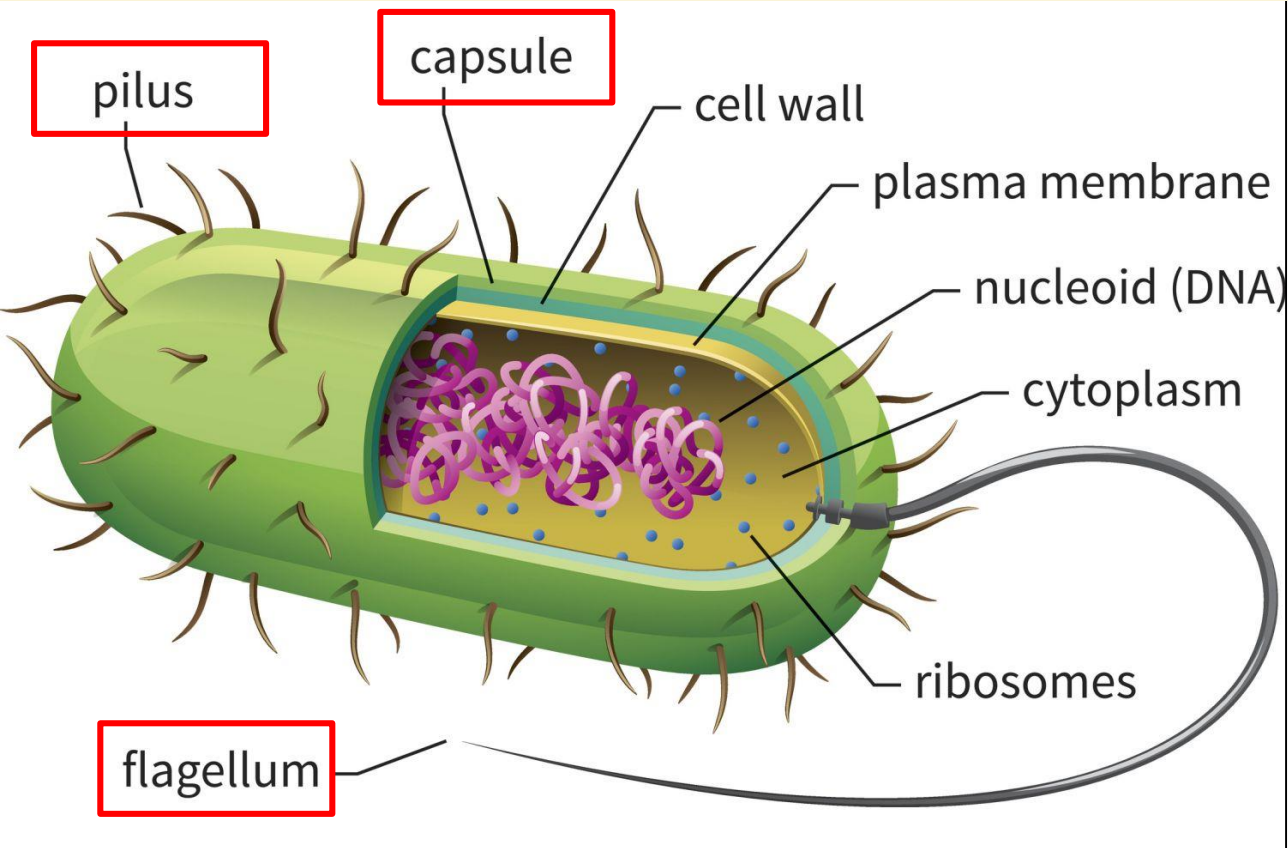
Characteristic	Prokaryote	Eukaryote
Nucleus	Absent	Present
Diameter of a typical cell	$\approx 1\mu\text{m}$	10–100 μm
Cytoskeleton	Absent	Present
Cytoplasmic organelles	Absent	Present
<u>DNA</u> content (base pairs)	1×10^6 to 5×10^6	1.5×10^7 to 5×10^9
Chromosomes	Single circular <u>DNA</u> molecule	Multiple linear <u>DNA</u> molecules

<https://www.youtube.com/watch?v=zZtcMBTQaS4>

<https://www.youtube.com/watch?v=Pxujitlv8wc>

Prokaryotes

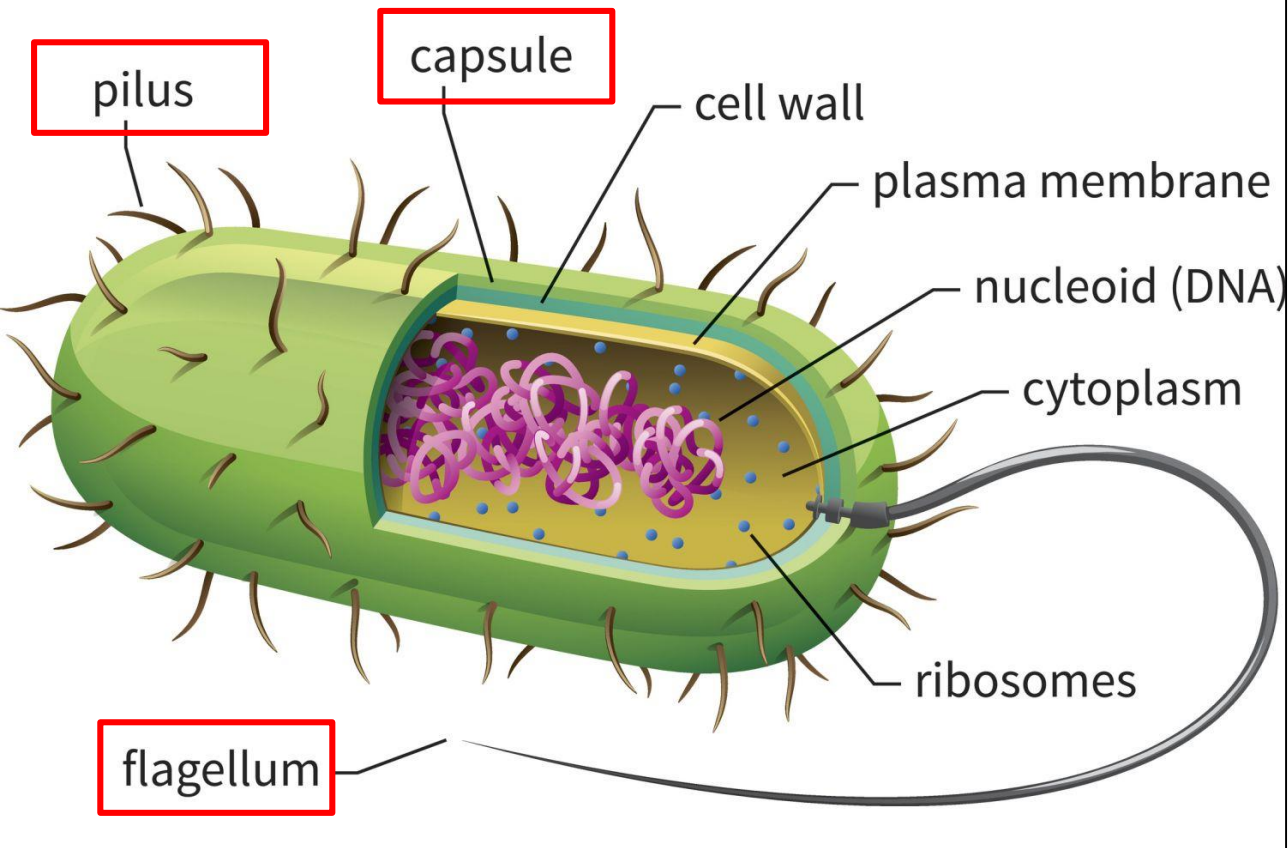
- ❑ It is a single-celled organism that lacks a membrane-bound nucleus (karyon), mitochondria or any other membrane-bound organelle.



- ❑ The **capsule** is a gelatinous layer covering the entire bacterium. It is composed of polysaccharide (long chains of carbohydrate molecules).
- ❑ The capsule—which can be found in both gram negative and gram-positive bacteria—is different from the second lipid membrane – bacterial outer membrane, which contains lipopolysaccharides and lipoproteins and is found only in gram-negative bacteria.
- ❑ The capsule is considered a virulence factor because it enhances the ability of bacteria to cause disease (e.g. prevents phagocytosis).
- ❑ The capsule can protect cells from engulfment by eukaryotic cells, such as macrophages.
- ❑ A capsule-specific antibody may be required for phagocytosis to occur. Capsules also contain water which protects the bacteria against desiccation.
- ❑ They also exclude bacterial viruses and most hydrophobic toxic materials such as detergents.
- ❑ Capsules also help cells adhere to surfaces.

Prokaryotes

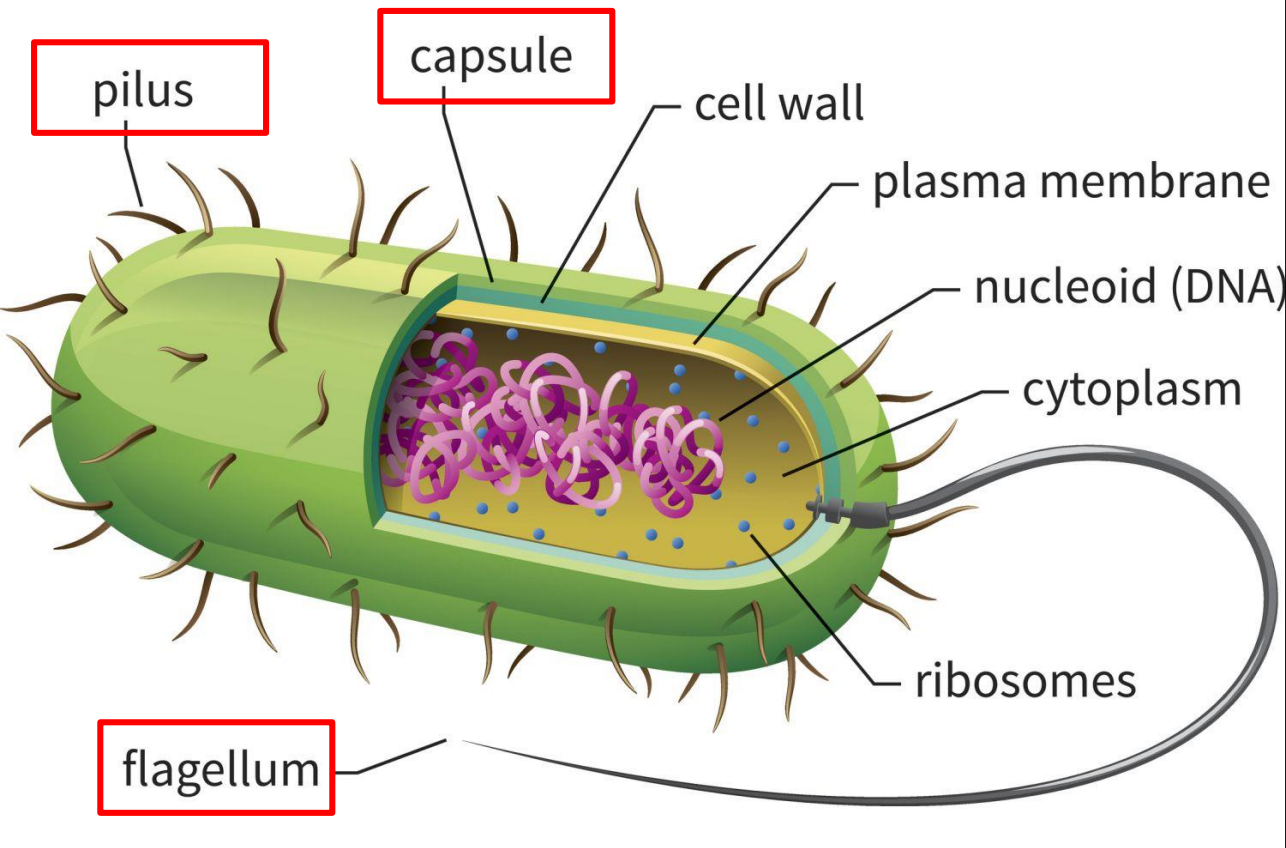
- ❑ It is a single-celled organism that lacks a membrane-bound nucleus (karyon), mitochondria or any other membrane-bound organelle.



- ❑ **Flagella** help move the bacteria toward nutrients and other attractants, a process called chemotaxis.
- ❑ The long filament, which acts as a propeller, is composed of many subunits of a single protein, flagellin, arranged in several intertwined chains.
- ❑ The energy for movement, the proton motive force, is provided by adenosine triphosphate (ATP), derived from the passage of ions across the membrane.

Prokaryotes

- ❑ It is a single-celled organism that lacks a membrane-bound nucleus (karyon), mitochondria or any other membrane-bound organelle.



- ❑ **Pili** are hair-like filaments that extend from the cell surface. They are shorter and straighter than flagella and are composed of subunits of pilin, a protein arranged in helical strands.
- ❑ Conjugative pili allow for the transfer of DNA between bacteria, in the process of bacterial conjugation. They are sometimes called "sex pili", in analogy to sexual reproduction, because they allow for the exchange of genes via the formation of "mating pairs".
- ❑ Pili are responsible for virulence in the pathogenic strains of many bacteria.

Various organelles in a human cell and its major function(s)



- 1. Cell membrane** – (is a membrane that separates the interior of all cells from the outside environment (the extracellular space) and protects the cell from its environment; acts as a barrier for cell contents. It consists of double phospholipid layer and monolayer of protein scattered around phospholipid layer. The other components in plasma membrane are cholesterol and glycoproteins).
- 2. Cytosol** – (fluid that contains organelles; represents the material outside the nucleus and inside the cell membrane)
- 3. Cytoskeleton** – (a network of protein structures that extend throughout the cytoplasm; provides the cell with an internal framework (e.g. microfilaments and microtubules))
- 4. Nucleus** – (control center of the cell; contains genetic material; It is the center of the cell because it contains genetic material (DNA). It consists of three main regions: the nuclear membrane, the nucleolus and chromatin.
Nuclear membrane: Nuclear membrane serves as a barrier of nucleus. It consists of a double phospholipid membrane and contains nuclear pores that allow for the exchange of material with the rest of the cell.
Chromatin: It is composed of DNA and protein scattered throughout the nucleus. Chromatin condenses to form chromosomes when the cell divides)
- 5. Nucleolus** – (functions as site of ribosome production; ribosomes then migrate to the cytoplasm through nuclear pores)
- 6. Ribosome** (little dots) – sites of protein synthesis in the cell.

Various organelles in a human cell and its major function(s)



7. **Rough endoplasmic reticulum** – (carry ribosomes that represent sites of protein synthesis)
8. **Smooth endoplasmic reticulum** – (function in cholesterol synthesis and breakdown, fat metabolism, and detoxification of drugs)
9. **Golgi apparatus** (or "Golgi body") – (modifies and packages proteins, secrete vesicles, plasma membrane components and lysosomes)
10. **Mitochondrion** – (powerhouse of the cell; generate ATP for cellular activities)
11. **Lysosomes** – (suicidal bag of the cell; contain enzymes that digest non-usable materials within the cell)
12. **Peroxisomes** – (detoxify harmful substances and break down free radicals; involved in catabolism of very long chain fatty acids)
13. **Centrioles** – (involved in the organization of mitotic spindle fibers during cell division and in the completion of cytokinesis)
14. **Centrosomes** – (serves as the main microtubule organizing center (MTOC) of the animal cell, as well as a regulator of cell-cycle progression; are composed of two centrioles arranged at right-angles to each other)
15. **Vesicles** – (involved in metabolism, transport, buoyancy control, and temporary storage of food and enzymes)

Important Links to see Videos



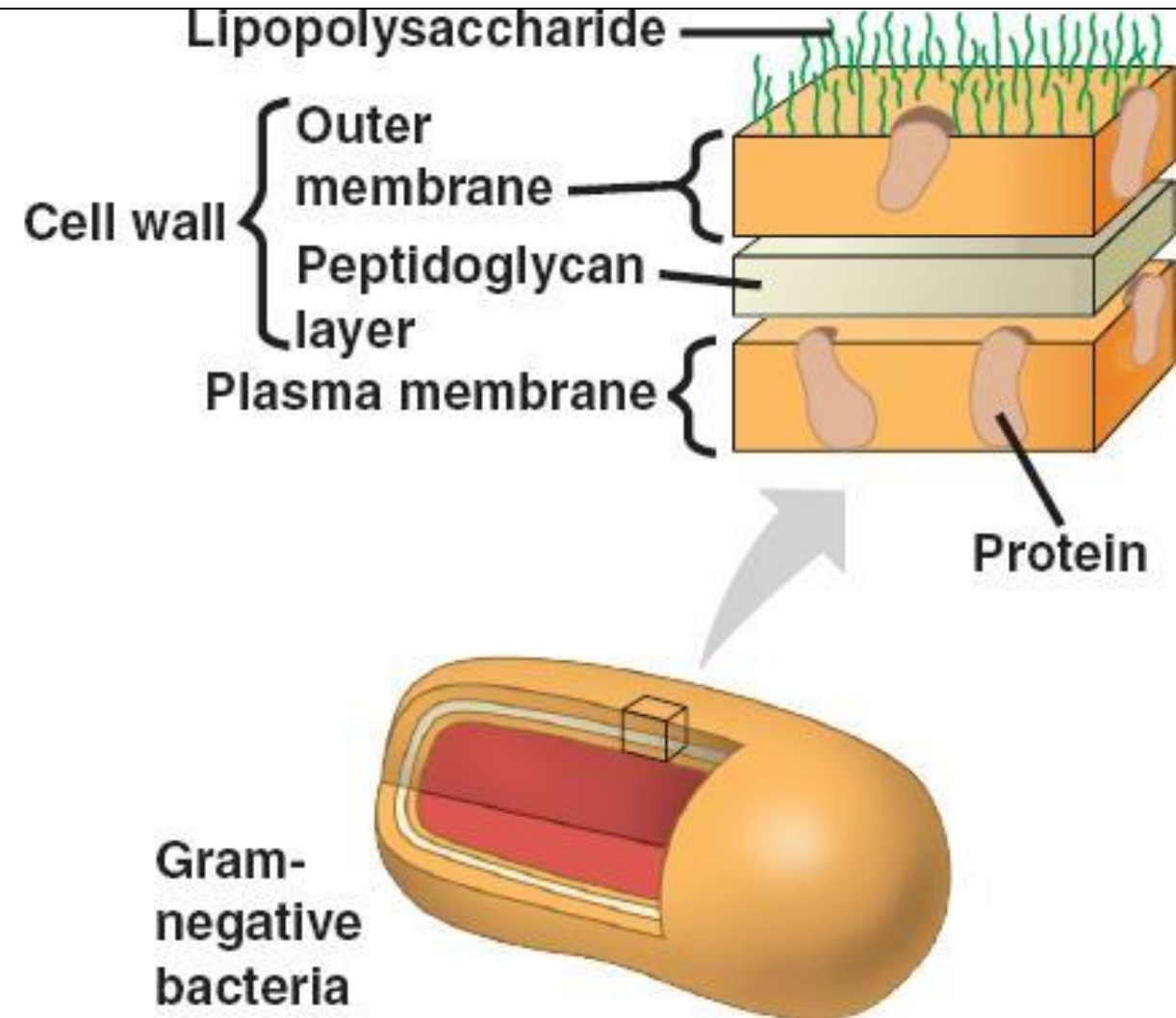
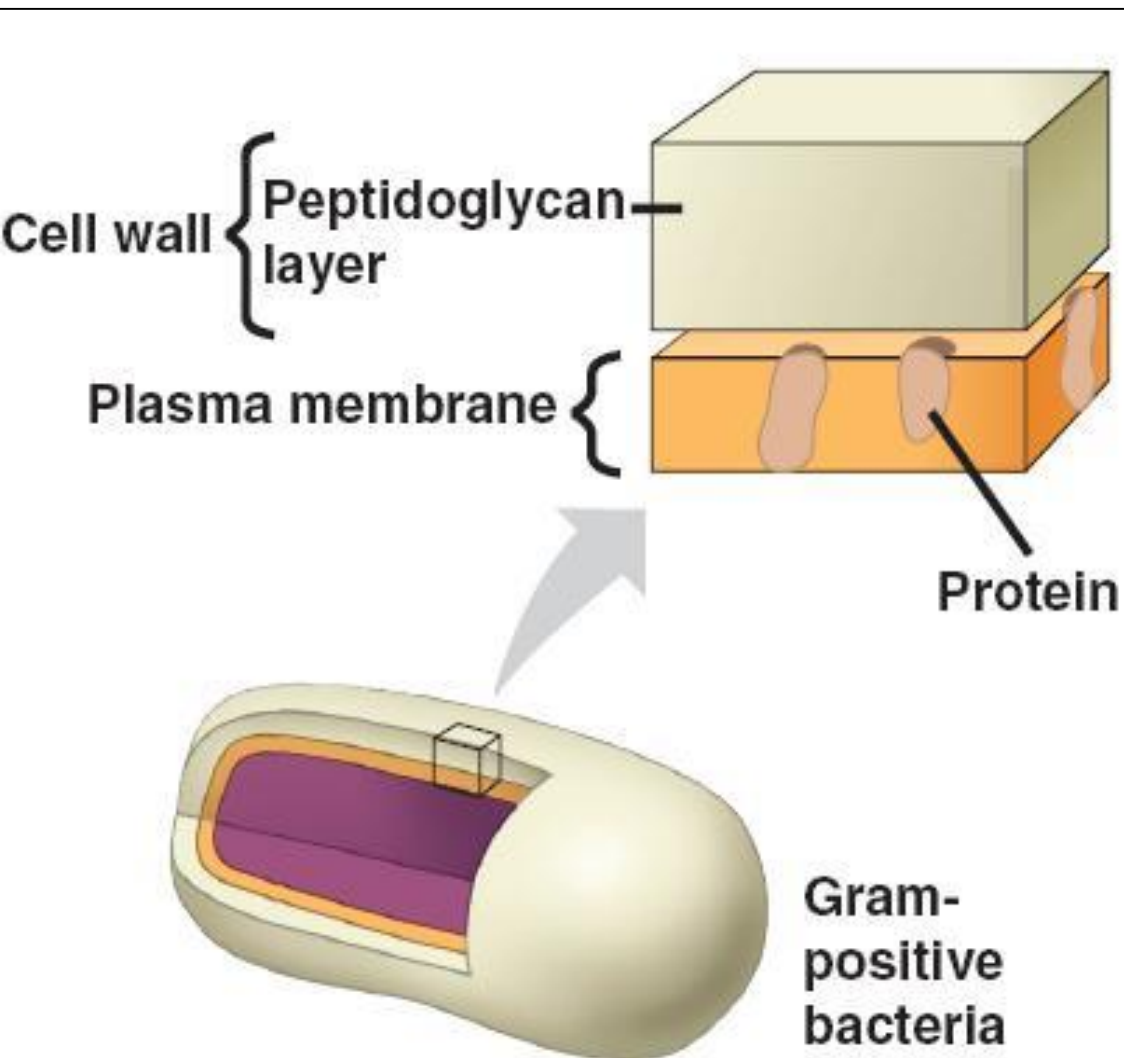
- ❑ **Overview of Cell Structure:**
- ❑ <https://www.youtube.com/watch?v=0xe1s65IH0w>
- ❑ <https://www.youtube.com/watch?v=RKmaq7jPnYM>

Cell wall

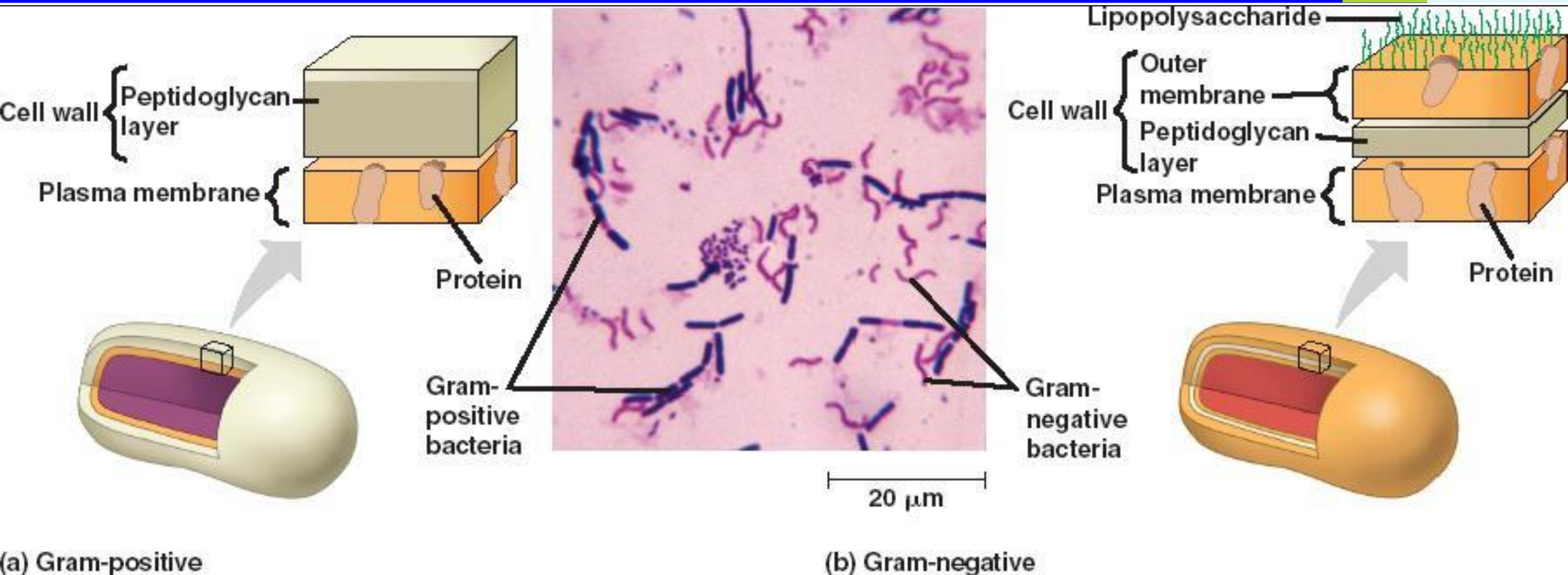


- ❑ The cell wall is a rigid and protective layer around the plasma membrane which provides mechanical support to the cell.
- ❑ Animal cells do not have a cell wall. They are present in most plant cells, fungi, bacteria, and some archaea.
- ❑ In plant cells, the cell wall is made up of cellulose (polysaccharide), hemicellulose (polysaccharide), pectin (polysaccharide), and protein.
- ❑ In many fungi, the cell wall is formed of chitin (polysaccharide).
- ❑ In bacteria, the cell wall contains protein-lipid-polysaccharide complexes. Bacterial cell walls are made of peptidoglycan (also called murein), which is made from polysaccharide chains cross-linked by peptides.
- ❑ The cell wall has many important functions in a cell including protection, structure, mechanical strength and support.
- ❑ **The cell wall protect organisms against viruses and other pathogens.**

Bacteria

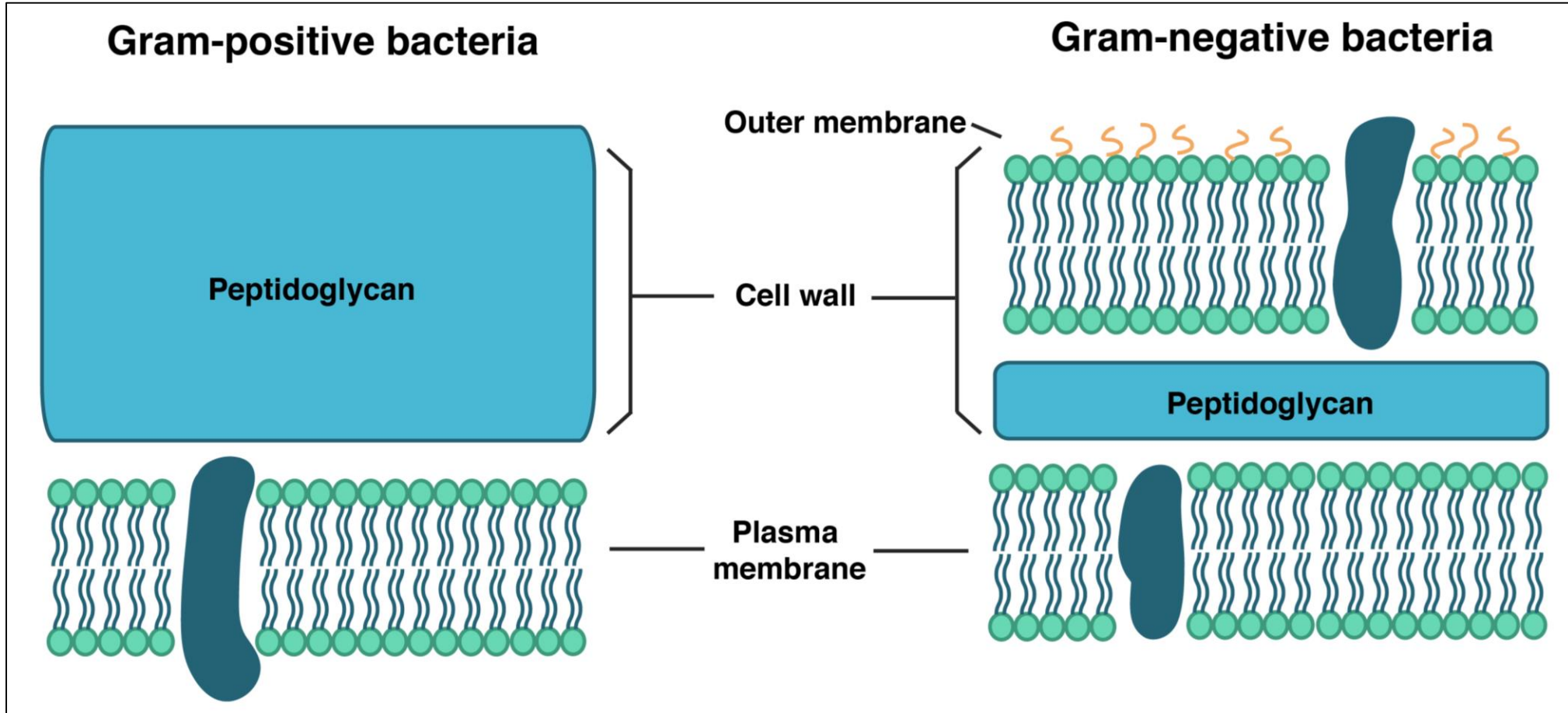


Bacteria



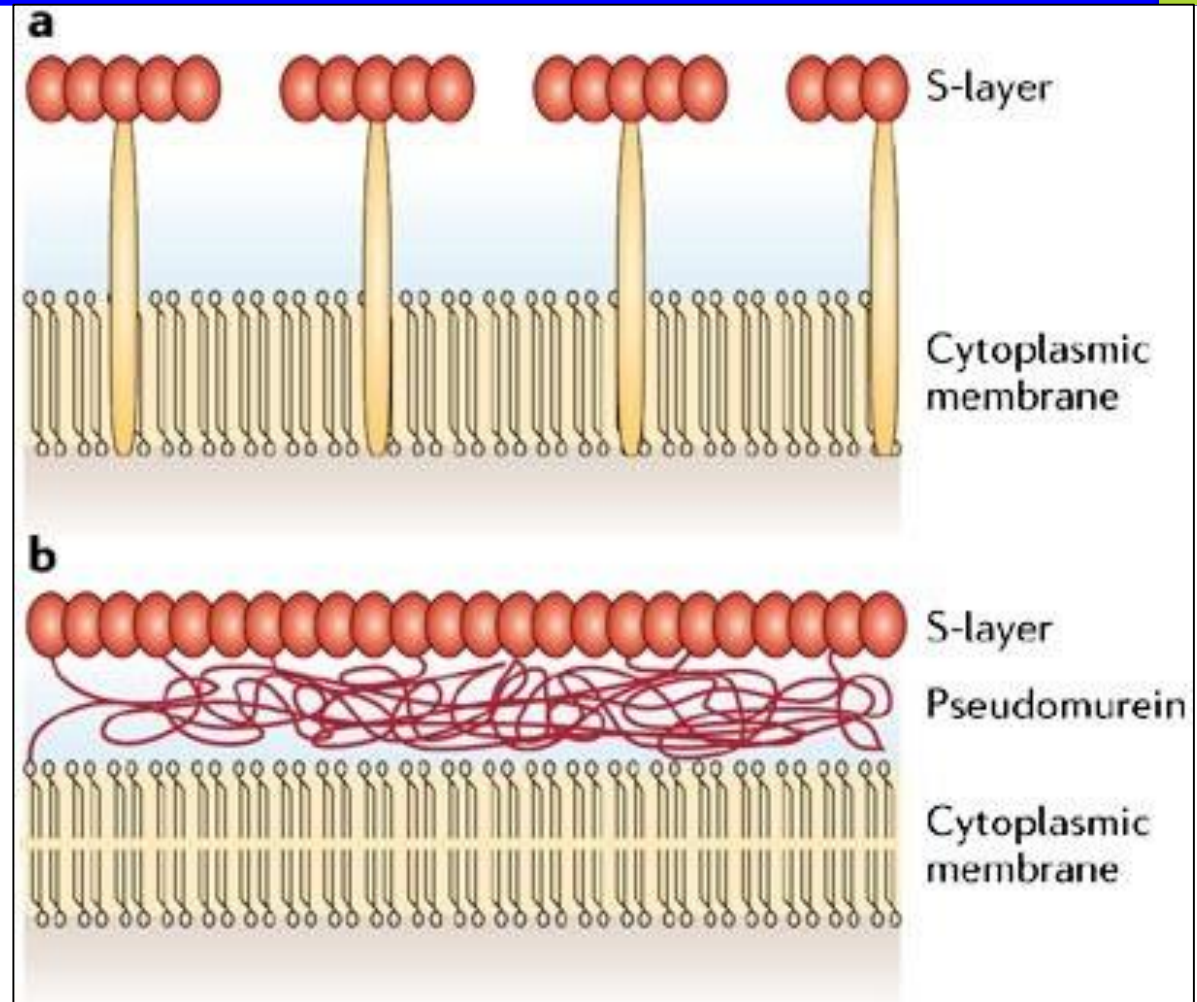
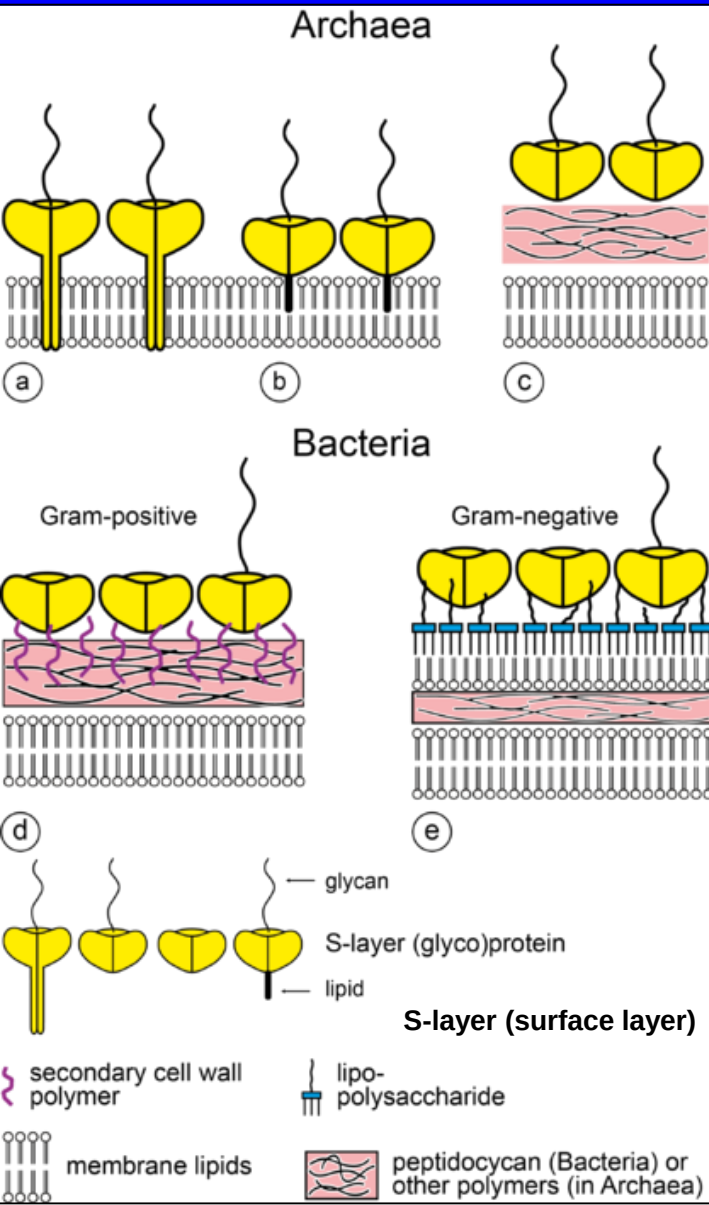
- ❑ Gram stain or Gram staining is a method of staining used to classify bacterial species into two large groups: Gram-positive bacteria and Gram-negative bacteria.
- ❑ Gram-positive bacteria have a thick mesh-like cell wall made of peptidoglycan (50–90% of cell envelope), and as a result are stained purple by crystal violet, whereas gram-negative bacteria have a thinner layer (10% of cell envelope), so do not retain the purple stain and are counter-stained pink by safranin.

Bacteria



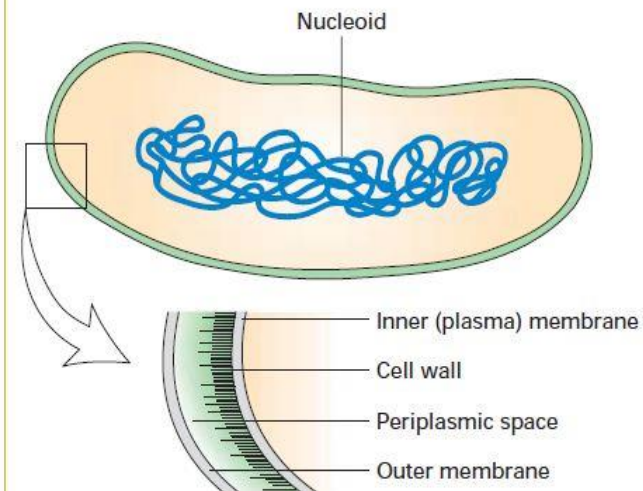
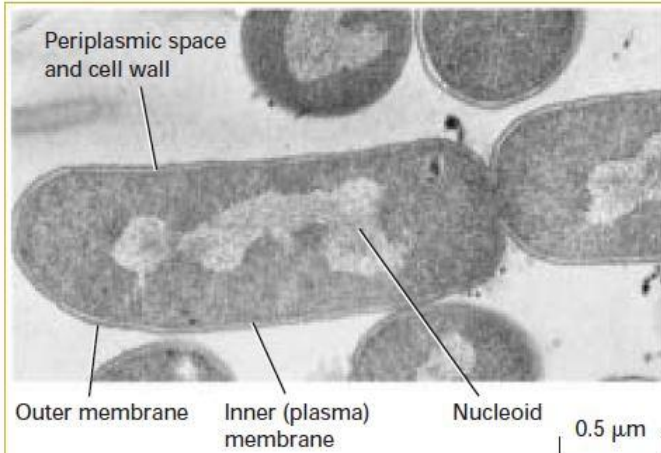
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Archaea vs Eubacteria (cell wall)



Prokaryotic cells have a simpler internal organization than eukaryotic cells

(a) Prokaryotic cell



(b) Eukaryotic cell

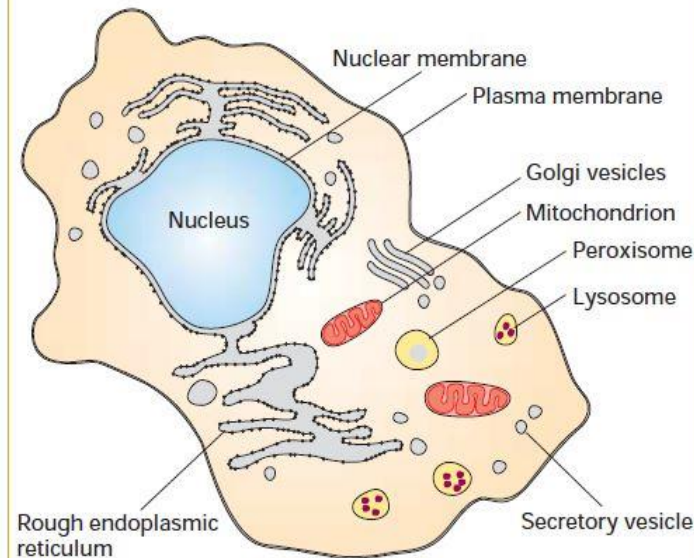
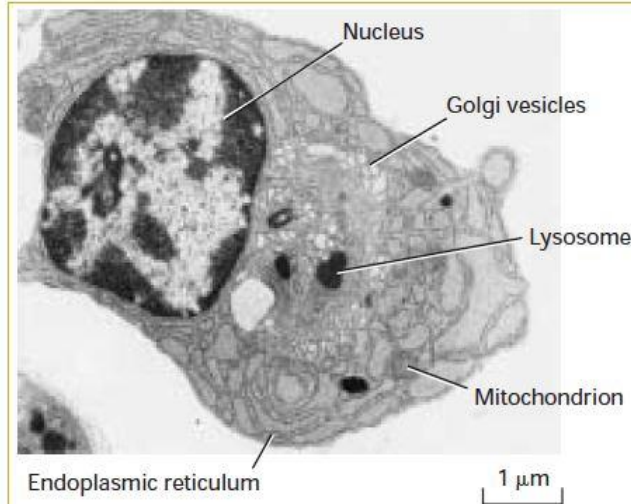


Figure 3.

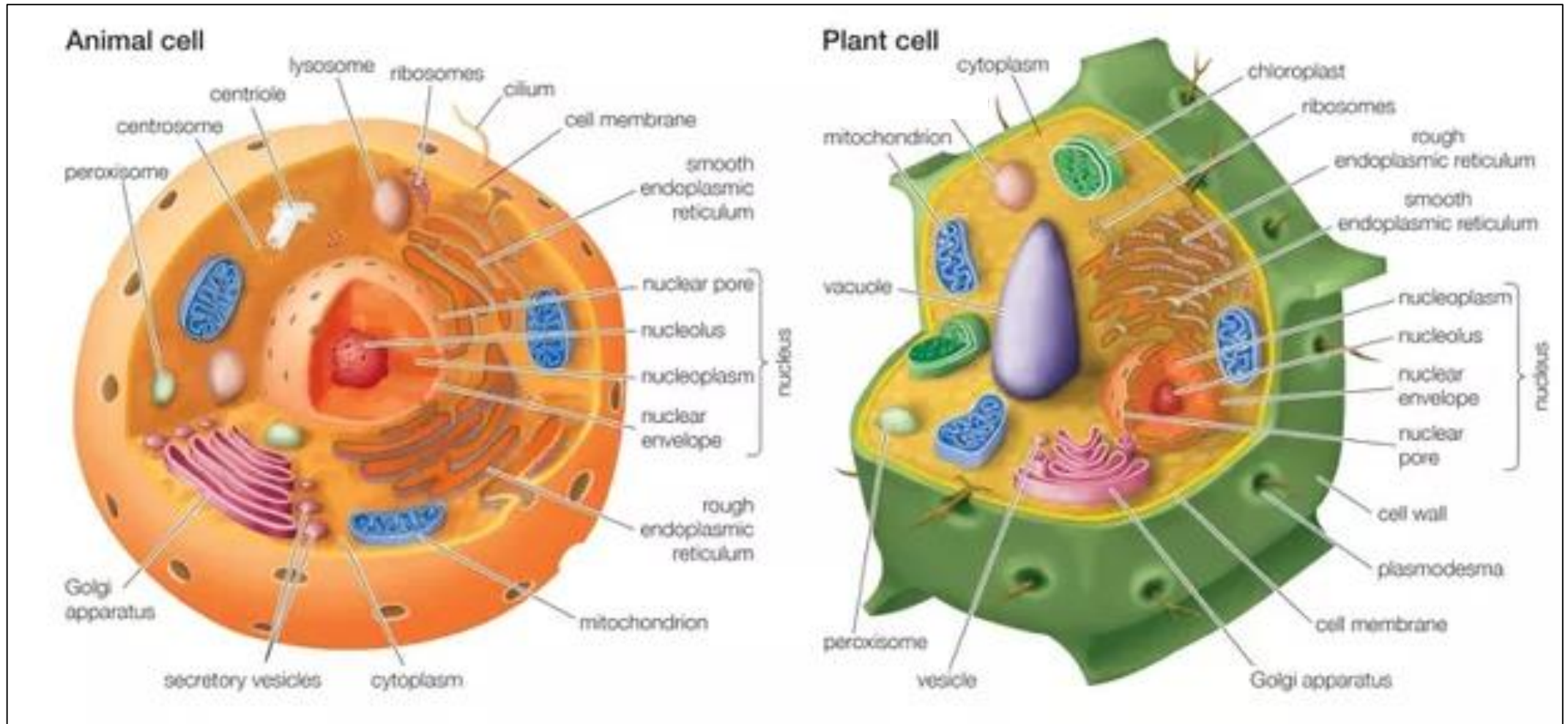
- a) Electron micrograph of a section of *Escherichia coli*, a common intestinal bacterium. **Nucleoid, consisting of DNA is not enclosed in a membrane.**
- b) Electron micrograph of White Blood Cell that secretes antibodies. **Only a single membrane (the plasma membrane) surrounds the cell, but the interior contains many membrane-bound organelles.**

An electron microscope is a microscope that uses a beam of accelerated electrons as a source of illumination. As the wavelength of an electron can be up to 100,000 times shorter than that of visible light photons, electron microscopes have a higher resolving power than light microscopes and can reveal the structure of smaller objects

**Credit: Molecular Biology of the Cell - Alberts*

Eukaryotes

❑ It is any organism whose cells contain a nucleus and other organelles enclosed within membranes.



Animal cells vs Plant cells



- ❑ Both plant and animal cells are eukaryotic, so they contain membrane-bound organelles like the Nucleus and Mitochondria.
- ❑ Animal and Plant cells, both are Eukaryotic Cells, yet there are some differences among them, and they do not look exactly the same or have all of the same organelles, since each has different needs.

For example, plant cells contain **Chloroplasts**, since they need to perform photosynthesis, but animal cells do not.

Animal cells vs Plant cells



- ❑ **Both animal and plant cells have mitochondria, but only plant cells have Chloroplasts.**
- ❑ Plants don't get their sugar from eating food, so they need to make **sugar from sunlight**. This process (**Photosynthesis**) takes place in chloroplast. Once the sugar is made, it is broken down by the mitochondria to make energy for the cell. Because animals get sugar from the food they eat, so they do not need chloroplasts, but just **Mitochondria**.
- ❑ **Both plant and animal cells have Vacuoles.** A plant cell contains a large, single vacuole that is used for storage and maintaining the shape of the cell. In contrast, animal cells have many, smaller vacuoles.
- ❑ **Plant cells have a cell wall, as well as a cell membrane. In plants, the cell wall surrounds the cell membrane.** This gives the plant cell its unique shape. Animal cells simply have a **Cell membrane**, but no Cell wall.
- ❑ **Centriole** is present only in **Animal cells** and in some **lower plant cells** e.g. mosses and ferns, but **not present in higher plant cells**. Plants do not have centrioles but they possess microtubules which acts just like a centriole that is it helps in the spindle fibre (aggregates of microtubules that move chromosomes during cell division) formation during cell division. A **Centrosome** comprises two microtubule rings known as **Centrioles**. Its main function is to organize the microtubules and provide a **structure** to the **cell**.

Animal cells vs Plant cells

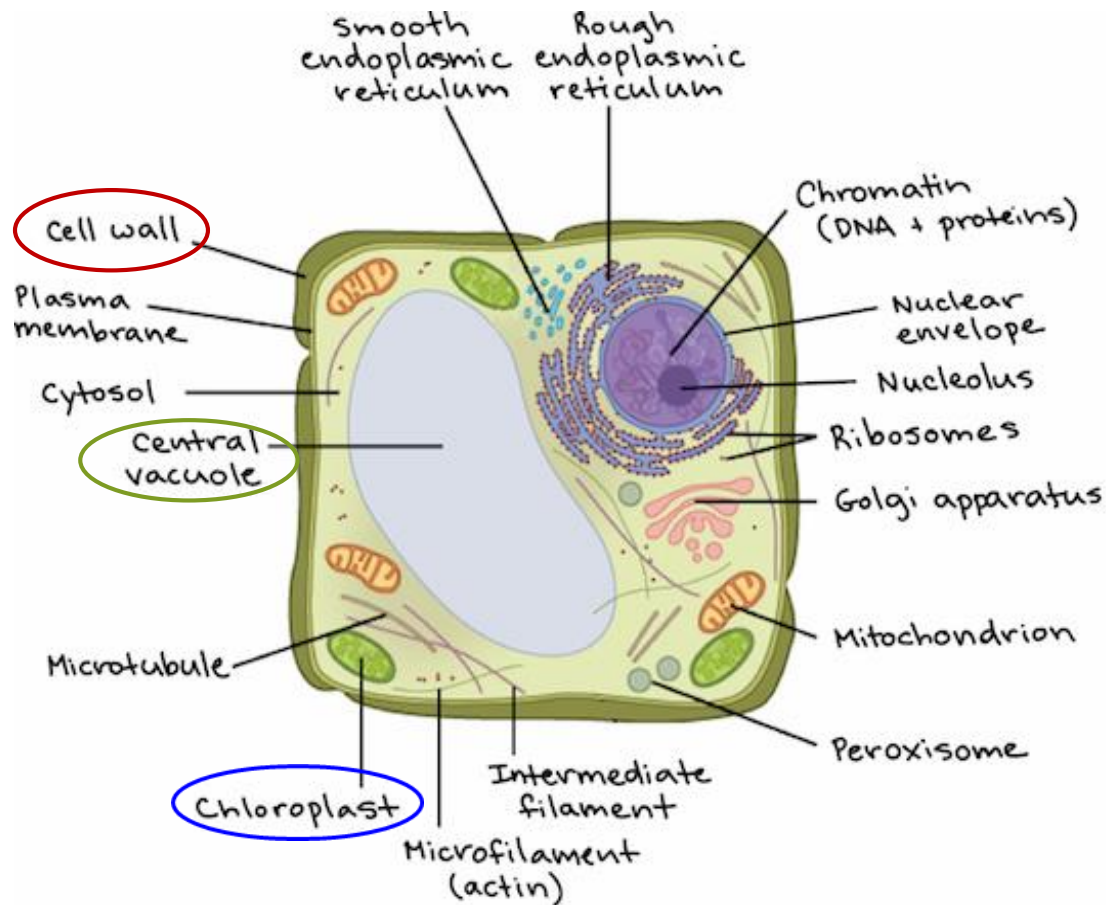


Figure 1a). Plant Cell

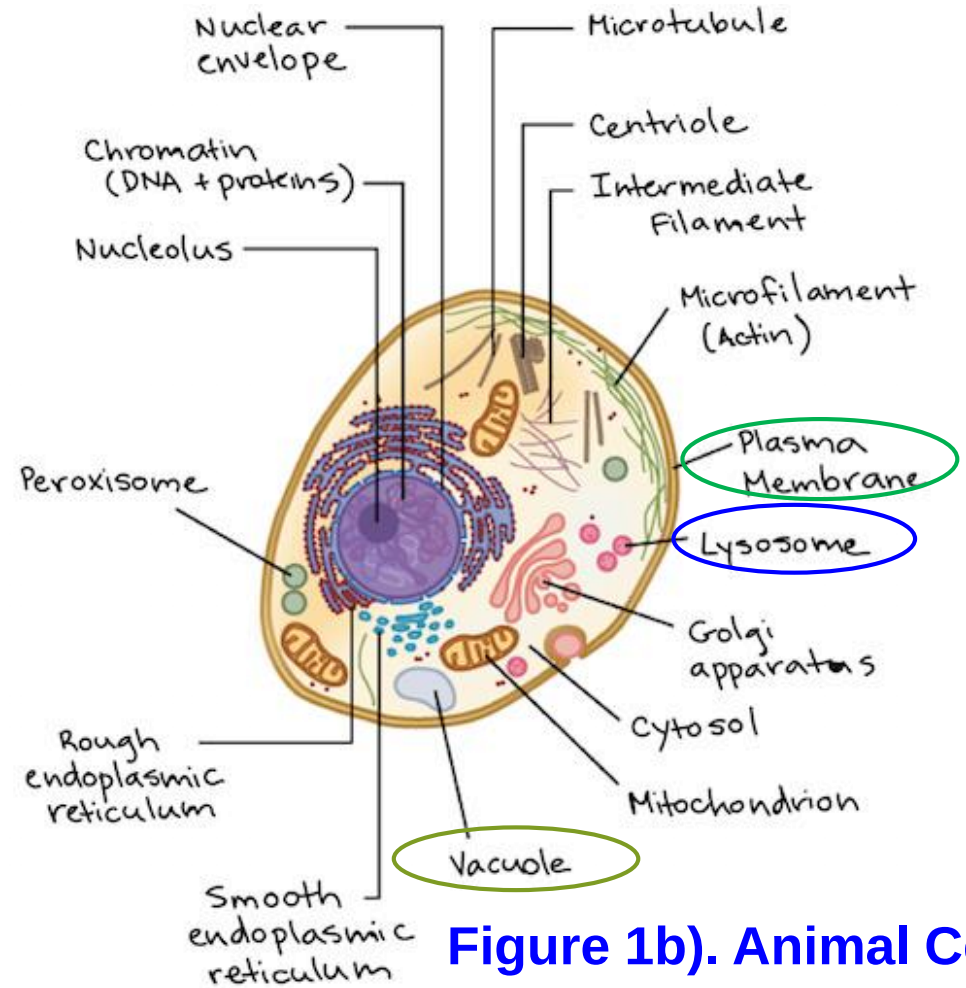


Figure 1b). Animal Cell

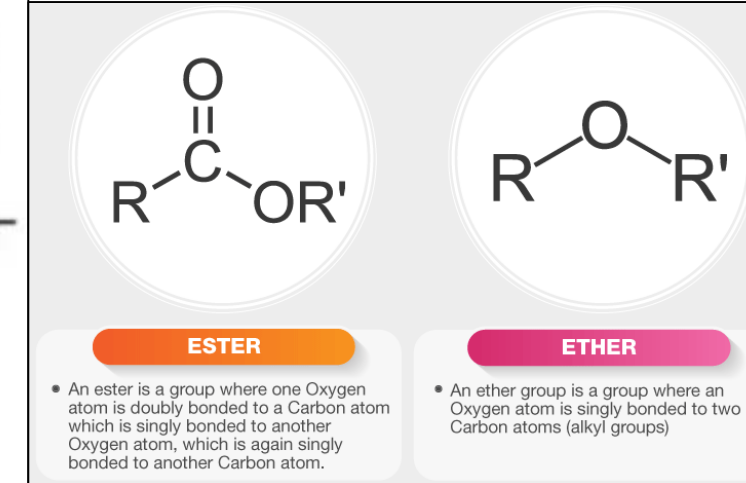
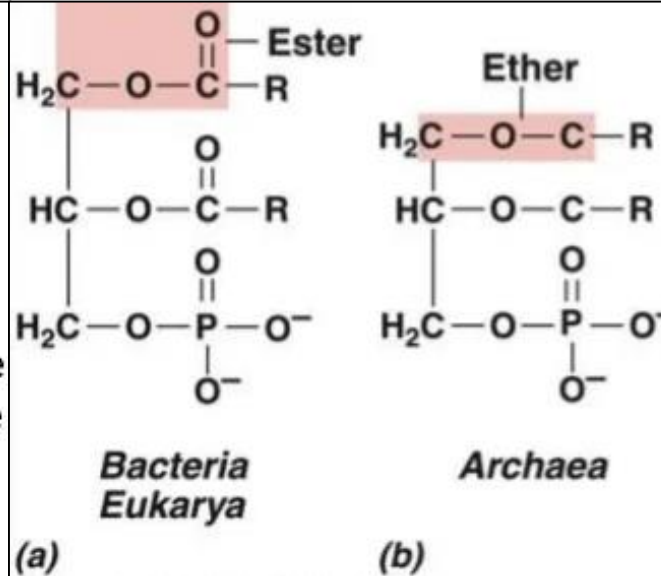
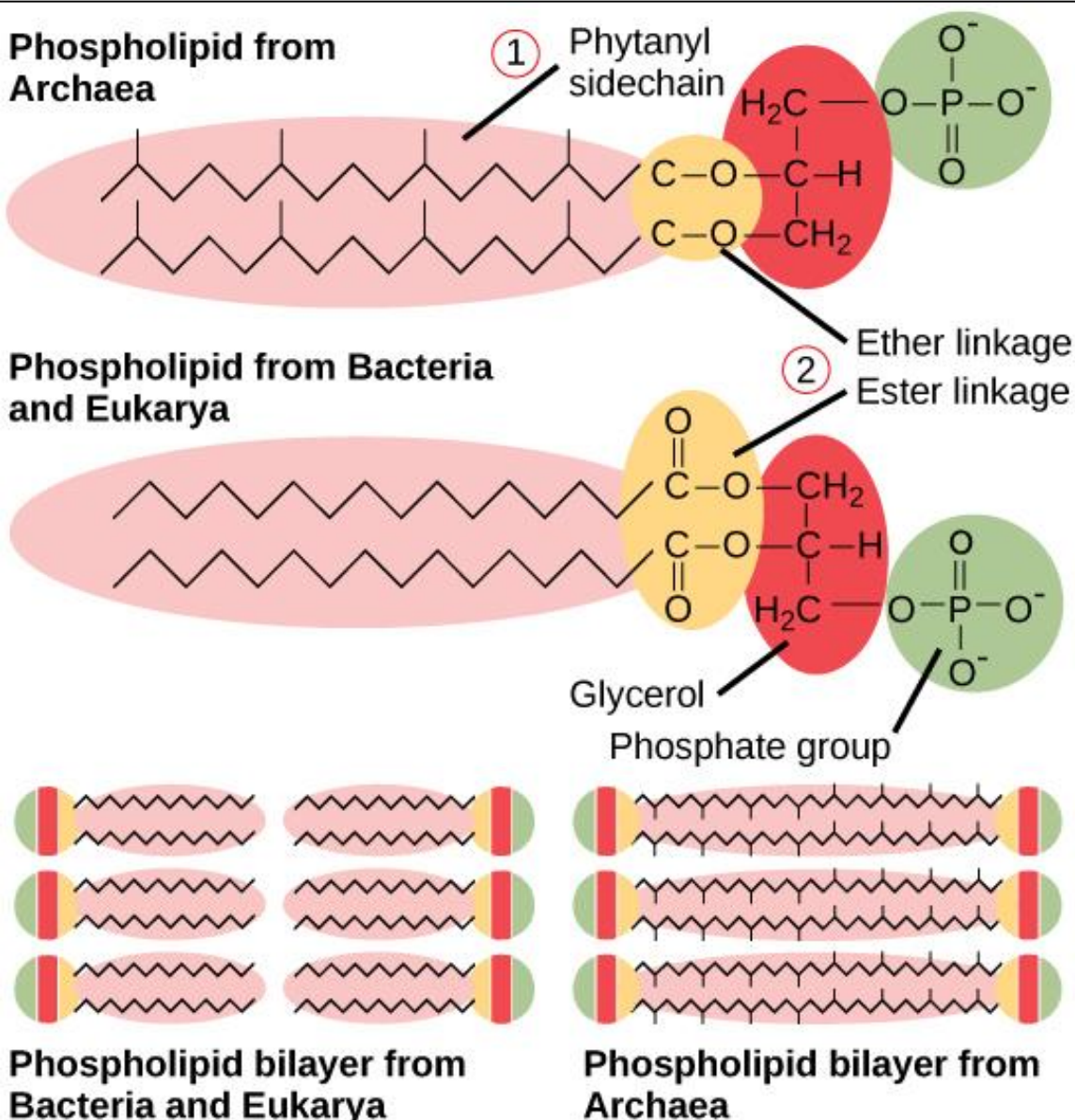
- Plant cell contains Chloroplast and its cell membrane is supported by rigid Cell wall. Whereas, an animal cell lacks chloroplast and cell wall. Instead, it contains only the Cell or Plasma membrane.
- Animal cell contains Lysosomes but Plant and Fungal cells do not contain lysosomes.
- Instead, they contain a large central Vacuole that performs some functions like lysosome.

Biological Membranes



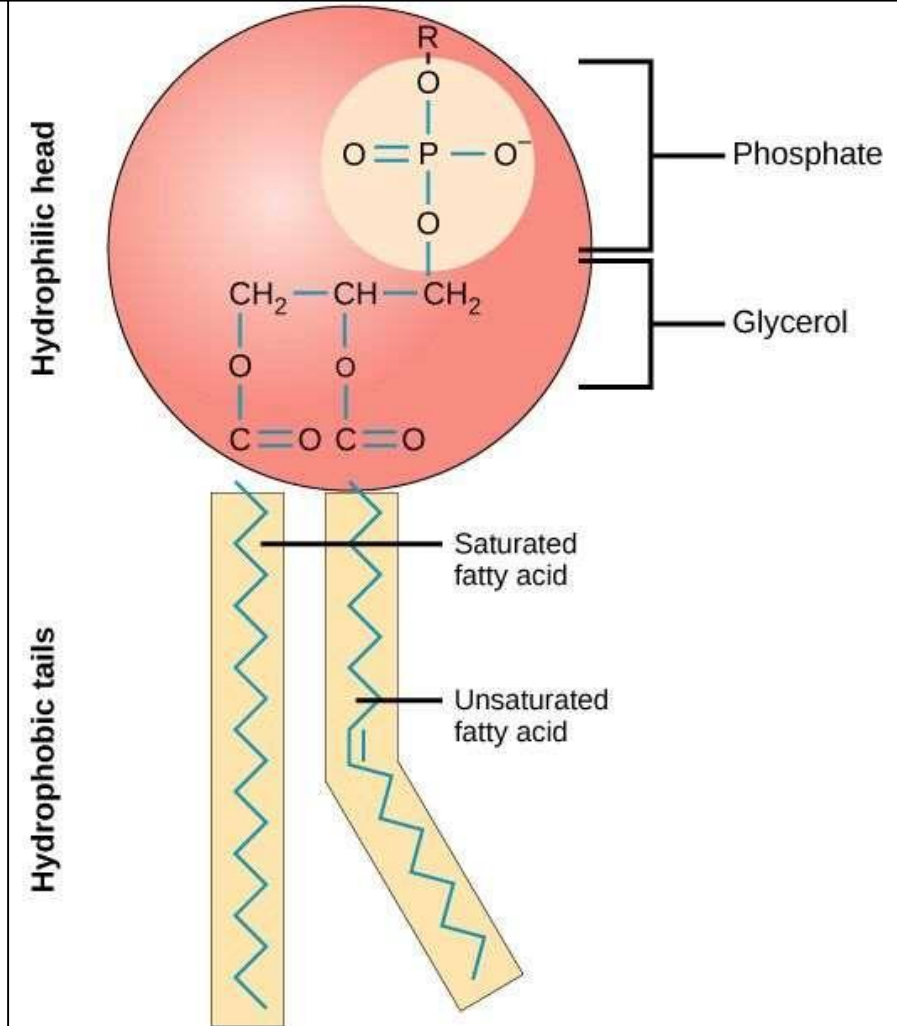
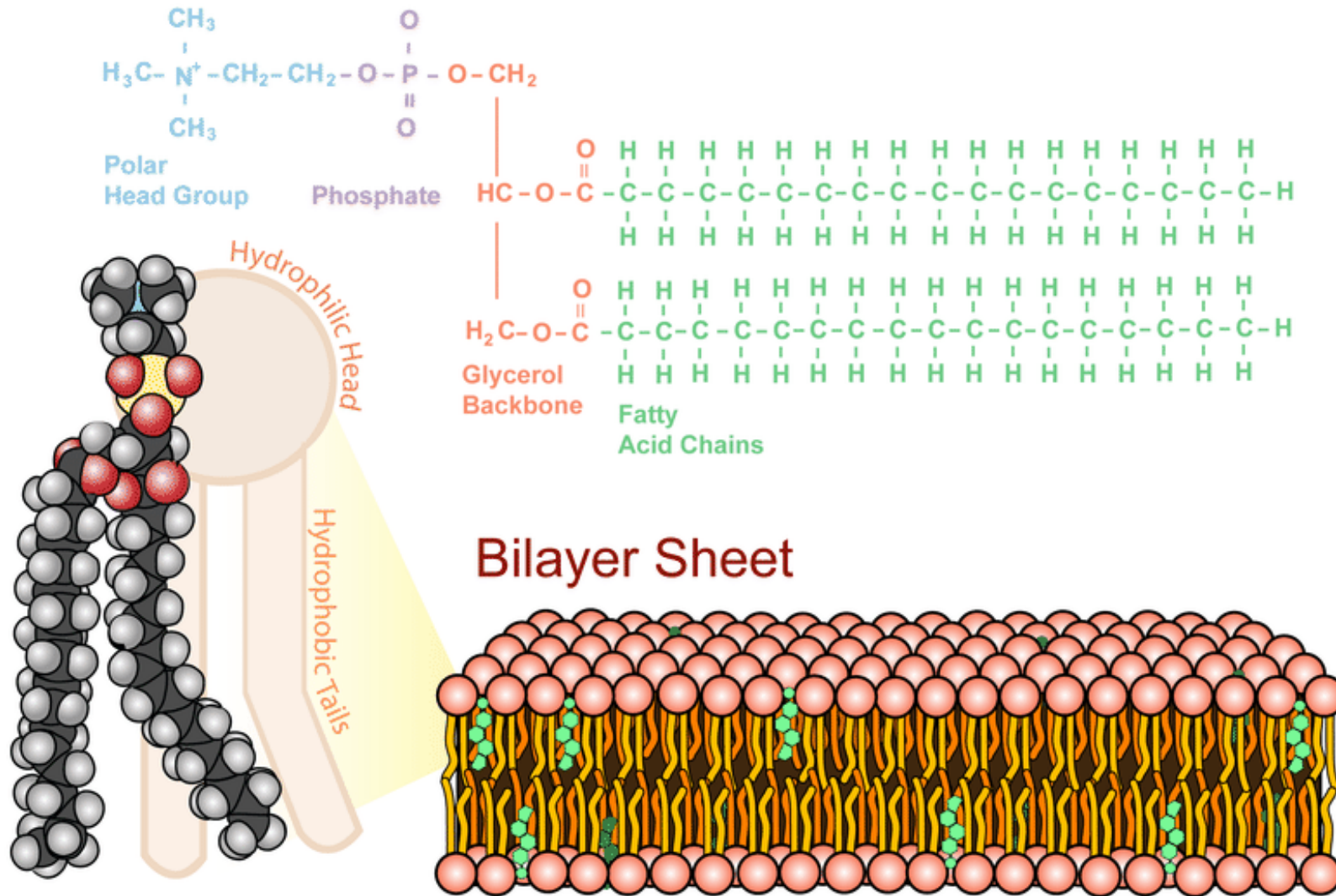
- ❑ **Membranes form boundaries both around cells and around subcellular organelles e.g. nucleus, mitochondria etc.**
- **They act as selectively, permeable barriers.**
- **All membranes contain two basic components: lipids and proteins. Some membranes also contain carbohydrate.**
- **The composition of lipid, protein and carbohydrate varies from one membrane to another.**
- ❑ **Membrane lipids**
- **Lipids are water insoluble oily organic substances that are soluble in non-aqueous solvents such as chloroform, ether, etc.**
- **In addition to being structural components of membranes, lipids serve as fuel molecules.**

Archaea vs Eubacteria vs Eukarya (cell membrane)



- ❑ Archaeol is one of the main core membrane lipids of archaea, one of the three domains of life.
- ❑ Standard archaeol is 2,3-di-O-phytanyl-sn-glycerol, with two phytanyl chains binding to the position of sn-2 and sn-3 of glycerol by ether bonds.
- ❑ Archaeol is generally composed by linking two phytanyl chains to the sn-2 and sn-3 positions of a glycerol molecule. The highly branched side chains are speculated to account for the very low permeability of archaeol-based membrane, which may be one of the key adaptations of archaea to extreme environment.
- ❑ One of the key features that distinguishes archaea from bacteria and eukarya is their membrane lipids, where archaeol plays an important role.
- ❑ Because of this, archaeol is also broadly used as a biomarker for ancient archaea.

Lipid bilayer



Lipid bilayer

Saturated Fat

meats, butter,
dairy products

solid at room
temperature

increase levels of
“**bad**” cholesterol
(low-density lipoprotein)

low-density lipoprotein
clogs arteries

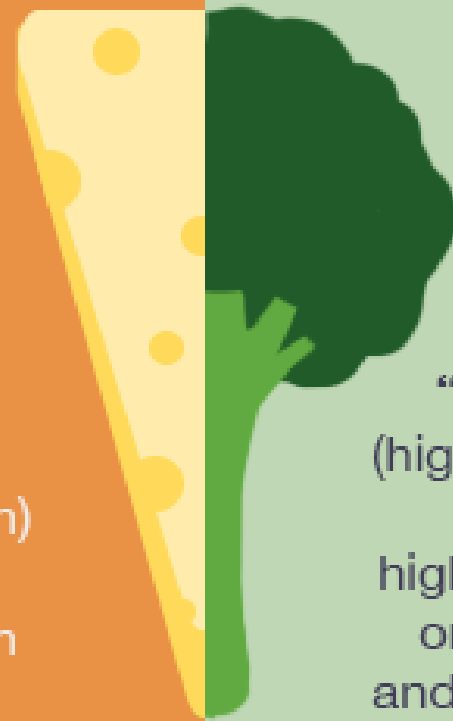
Unsaturated Fat

vegetable oils

liquid at room
temperature

increase levels of
“**good**” cholesterol
(high-density lipoprotein)

high-density lipoprotein,
or HDL, “grabs” LDL
and escorts it to the liver
where **LDL is broken down**
and eventually removed
from the body



Important Links to see Videos



- ❑ **Overview of Cell Membrane:**
- ❑ <https://www.youtube.com/watch?v=KGTcgShKJpw>
- ❑ <https://www.youtube.com/watch?v=fJfTDc3WzQ8>
- ❑ <https://www.youtube.com/watch?v=RT61MUjogRo>
- ❑ <https://www.youtube.com/watch?v=FYwB1EETakY>
- ❑ <https://www.youtube.com/watch?v=3k7Y6NcXe6s>

Organelles of the Eukaryotic Cells



- The Organelles are small, specialized structures in cells which perform specific functions.
- The cell organelles are of three types.

Those containing a, i) Double membrane, ii) Single membrane or iii) No membrane.

1. Double membrane bound cell organelles

1.1 Nucleus

1.2 Mitochondria

1.3 Chloroplast

2. Single membrane bound cell organelles

2.1 Endoplasmic reticulum

2.2 Golgi apparatus

2.3 Vacuole

2.4 Lysosomes and endosomes

2.5 Peroxisomes

3. Cell organelles with out the membrane

3.1 Ribosomes

3.2 Centrosomes (Centriole)

3.3 Cytoskeleton (Microtubules, actin filaments and intermediate filaments)

1.1 Nucleus

- The nucleus, the largest organelle in animal cells and is surrounded by **two membranes (outer and inner nuclear membrane)**, each one a phospholipid bilayer containing different types of proteins (**Figure 2**).
- The two nuclear membranes appear to fuse at **nuclear pores** (**Figure 2**), the ring-like complexes through which material moves between the nucleus and cytosol or cytoplasm.
- **Nuclear lamina** along the surface of the inner membrane is composed of proteins called **Lamins** which gives the shape to nucleus.
- In a growing or differentiating cell, the nucleus is metabolically active engaged in replicating DNA and synthesizing RNA (rRNA, tRNA and mRNA).

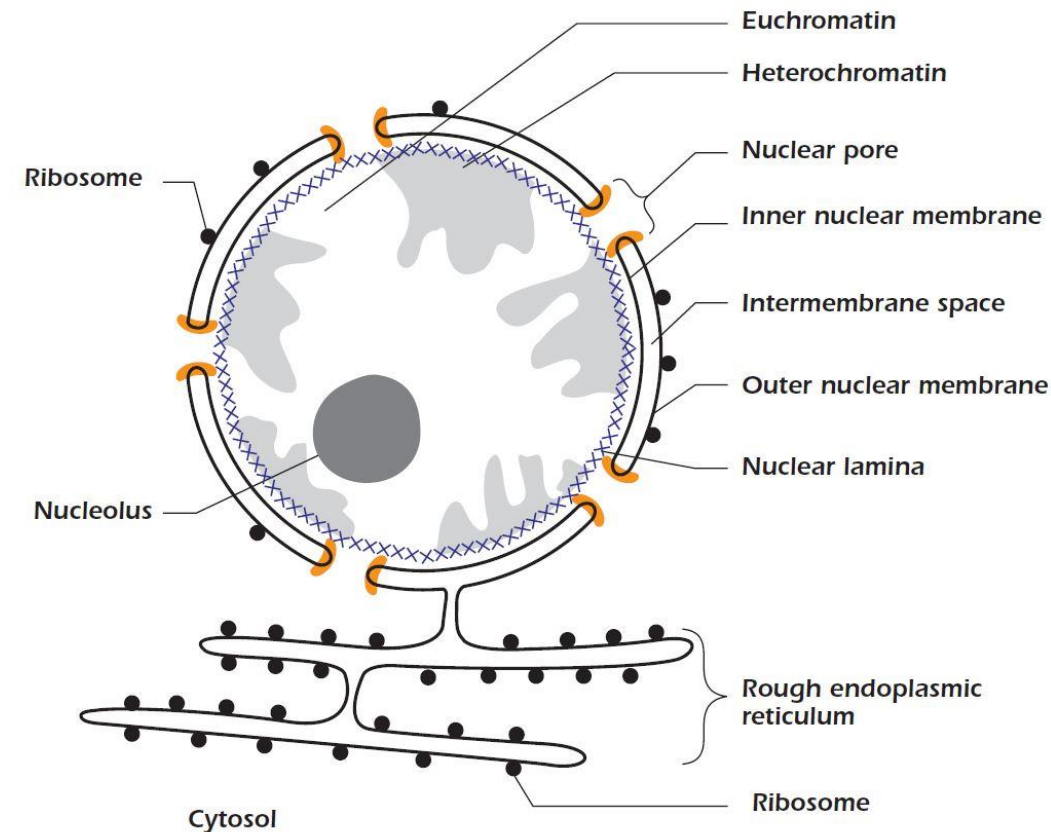


Figure 2. Nucleus and the relationship of its membranes with Endoplasmic Reticulum.

1.1 Nucleus

- In the electron microscope, the non-nucleolus region of the nucleus, called the **Nucleoplasm**, can be seen to have dark- and light-staining areas.
- The Dark areas, which are often associated with the nuclear membrane, contain condensed DNA (inactive for transcription), are called **Heterochromatin**.
- The Light areas containing the loose chromatin (DNA and protein) structure (active for transcription) are called **Euchromatin**.
- Most of the cell's **ribosomal RNA** is synthesized in the **Nucleolus**, present as a subcompartment in the Nucleus, that is not bound by any membrane.

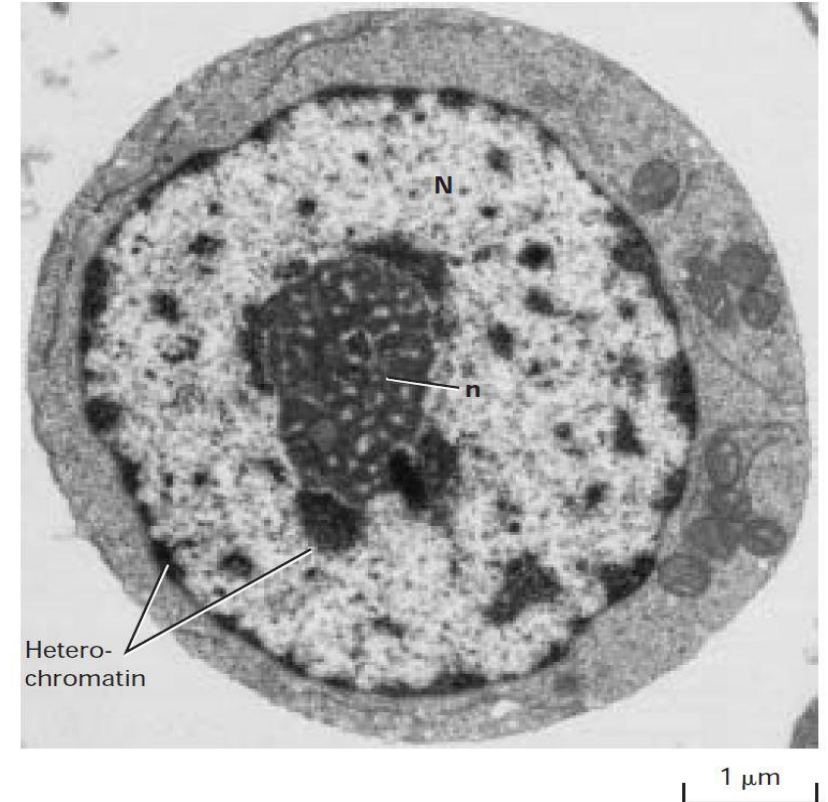


Figure 3. Electron micrograph of a thin section of a **Bone marrow** stem cell. **The nucleolus (n)** is a sub-compartment of nucleus (N) and is not surrounded by a membrane. **Dark stained areas in nucleus outside the nucleolus are heterochromatin.**



Important Links to see Videos



- ❑ **Nucleus:**
- ❑ <https://www.youtube.com/watch?v=UYj5aofYhpo>

1.2 Mitochondria: Structure

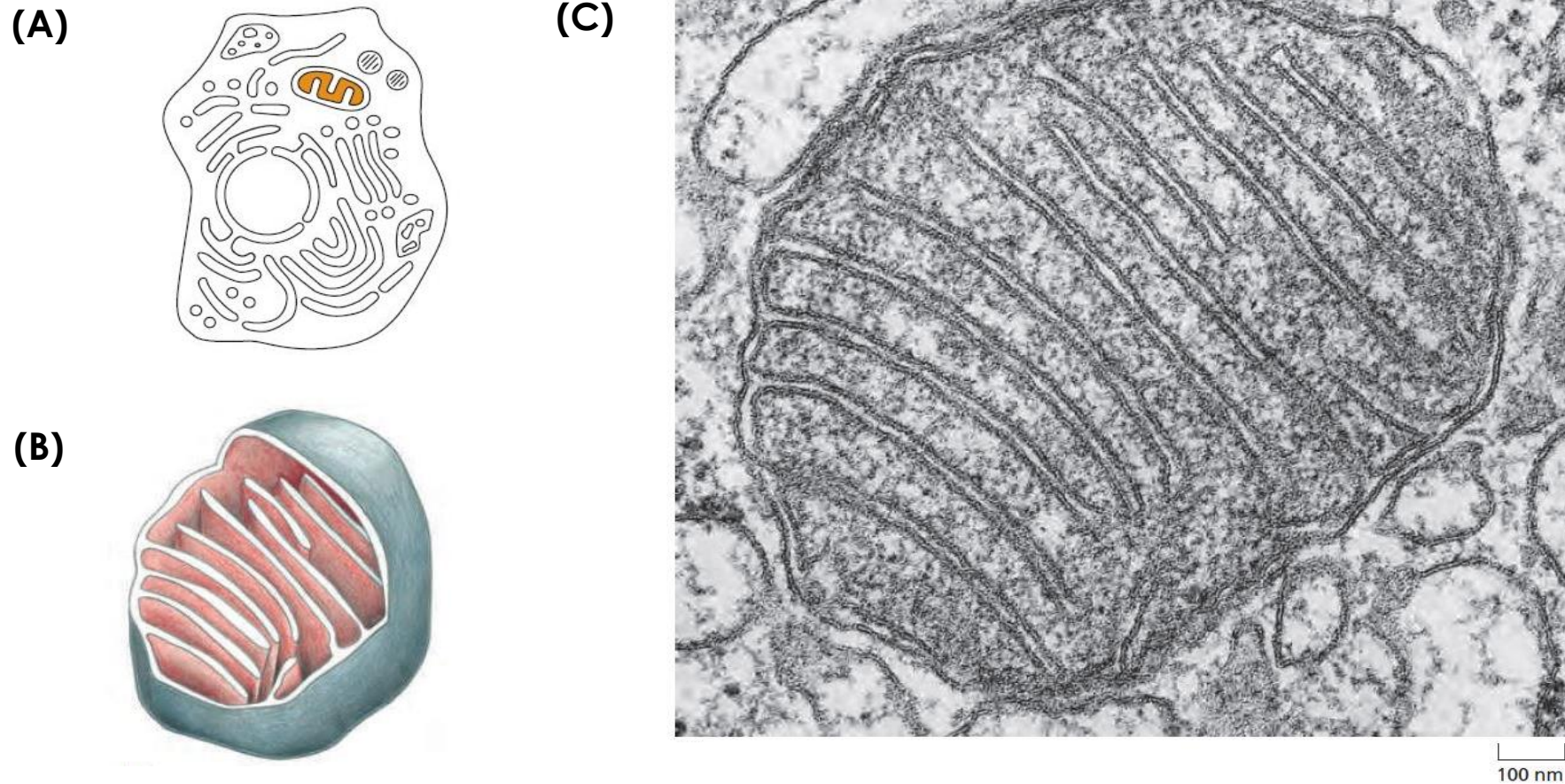


Figure 4. A) Schematic of Eukaryotic cell.

B) Drawing of a Mitochondrion with a part cut away, showing the 3-dimensional structure.

C) Cross-section of a Mitochondrion, as seen through image by Transmission Electron Microscope.

1.2 Mitochondria, The Power Plants of a cell

- Most eukaryotic cells contain many Mitochondria, which occupy up to 25% of the volume of the Cytoplasm.
- Mitochondrion (singular) is the main site for ATP synthesis during aerobic metabolism and is smaller in size than nucleus, vacuoles or chloroplasts.
- The two membranes (outer and inner) that bound a mitochondrion differ in composition and function.
- The surface area of the inner membrane is greatly increased by a large number of infoldings called **Cristae**, that protrude into the **Central space or Matrix** (Figure 5).

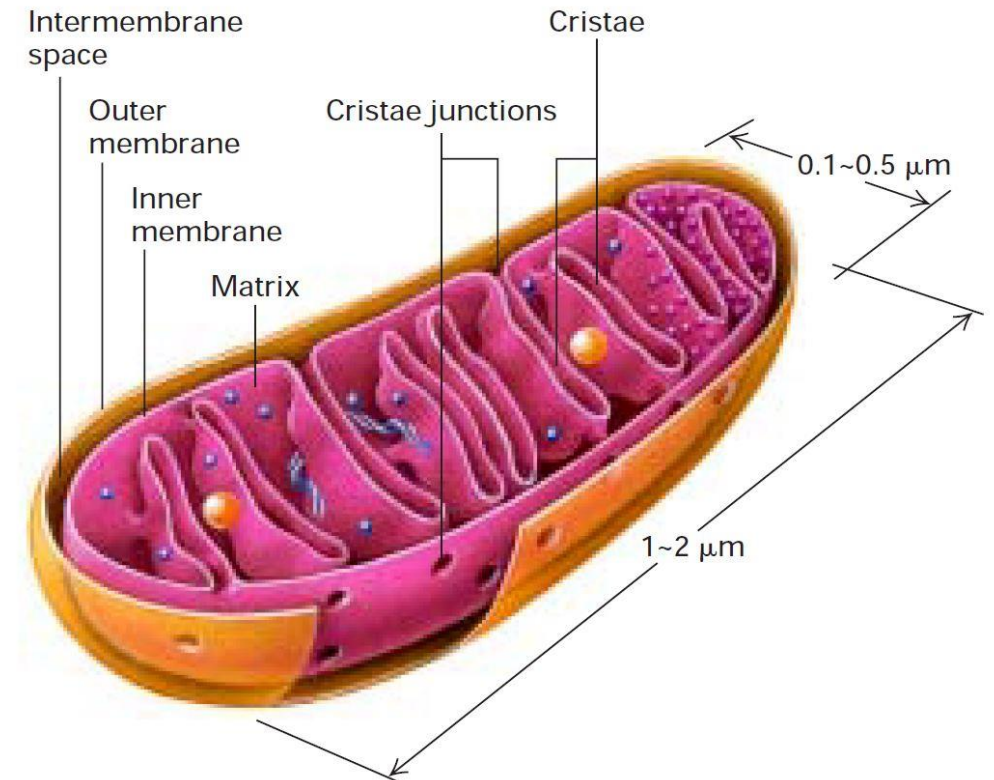
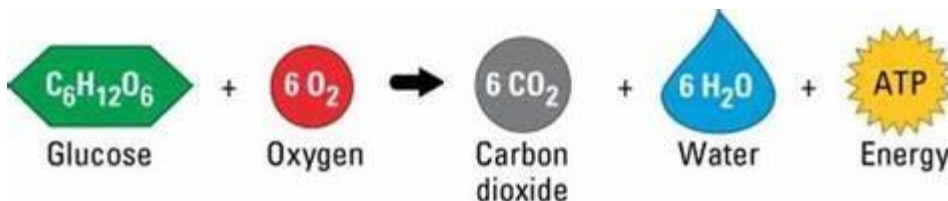
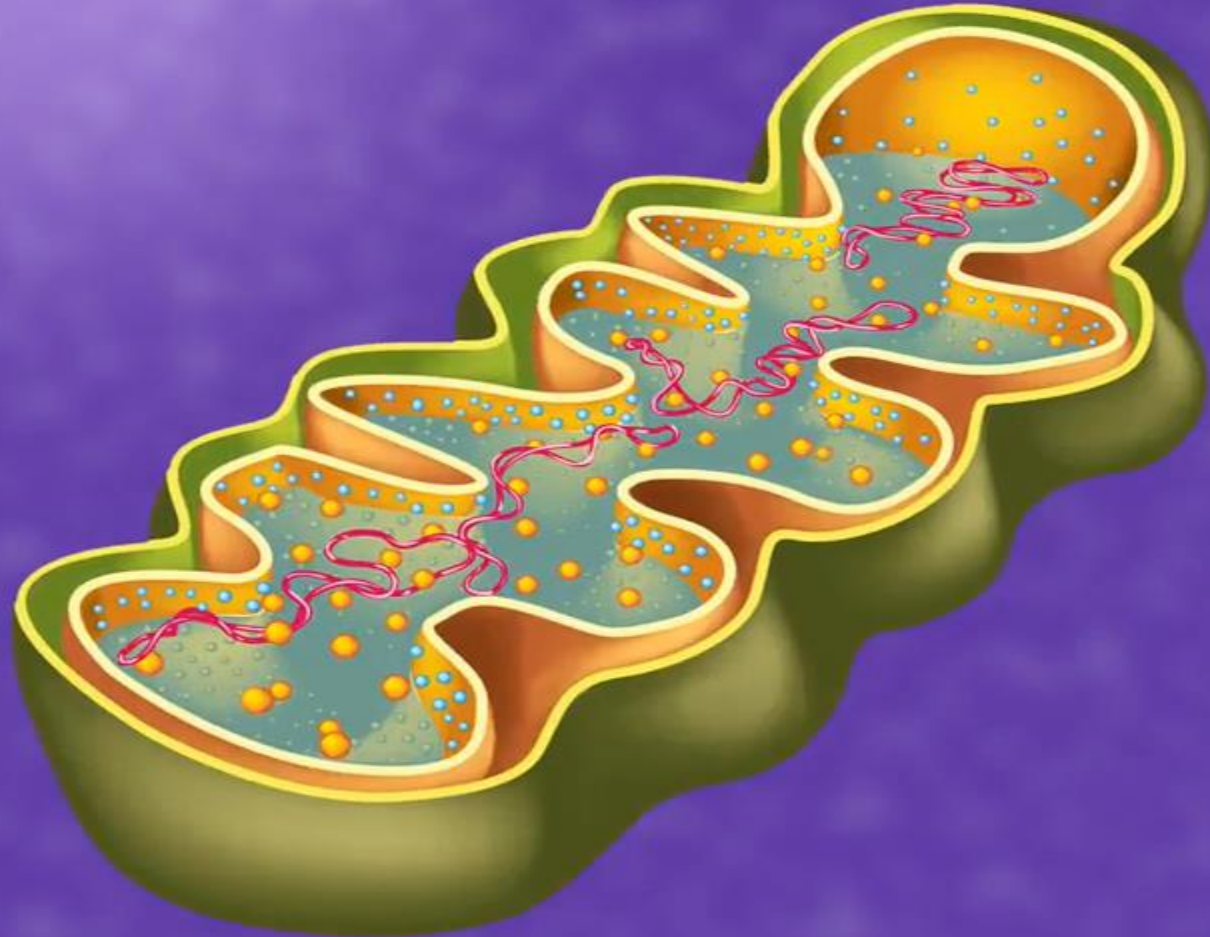


Figure 5. Schematic diagram of Mitochondrion showing membranes and compartments.

*Credit: Molecular Cell Biology-Lodish



STRUCTURE



How Mitochondria Produce Energy?



cell

Important Links to see Videos



- ❑ **Mitochondria:**
- ❑ <https://www.youtube.com/watch?v=c4JsEBI9u6I>
- ❑ <https://www.youtube.com/watch?v=39HTpUG1MwQ>
- ❑ <https://www.youtube.com/watch?v=rdF3mnyS1p0>

1.3 Chloroplasts

Chloroplasts are the largest and most characteristic organelles in the cells of plants and green algae after the Vacuoles.

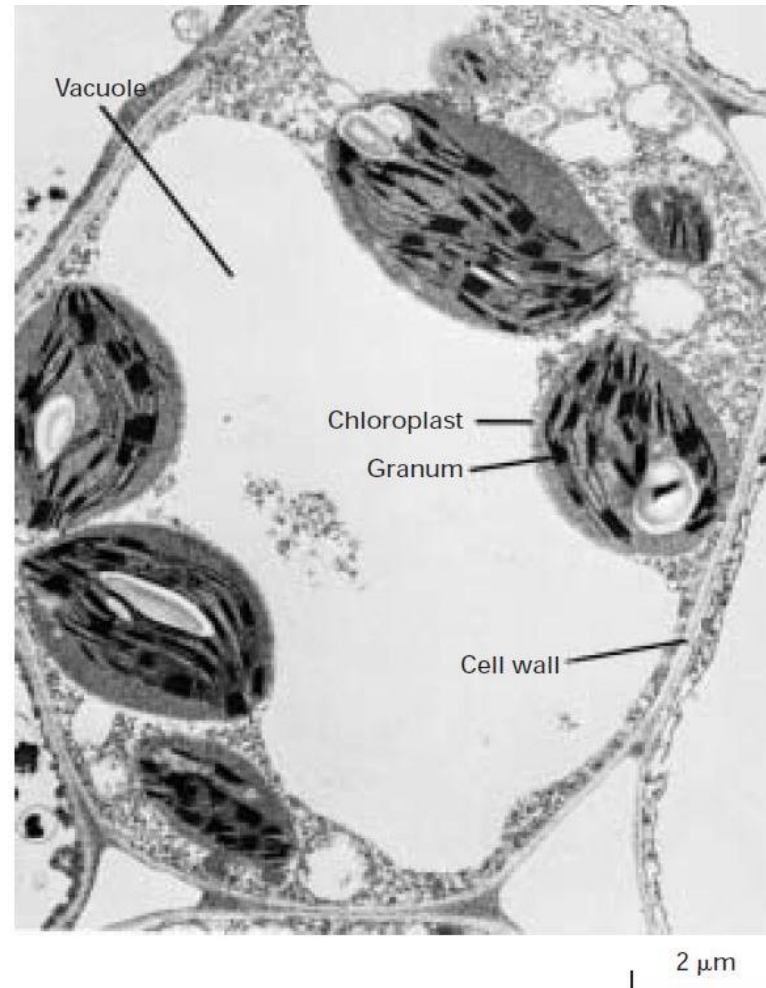


Figure 6. Chloroplast of Plant cell as seen by an image from Transmission Electron Microscope (TEM).

1.3 Chloroplasts

Chloroplasts are $\sim 10\text{ }\mu\text{m}$ long and $0.5 - 2.0\text{ }\mu\text{m}$ thick, but vary in size and shape in different cells, especially in algae.

Chloroplasts capture energy from sunlight in Plant cells and in single-celled Eukaryotes such as, *Volvox* (a Green alga).

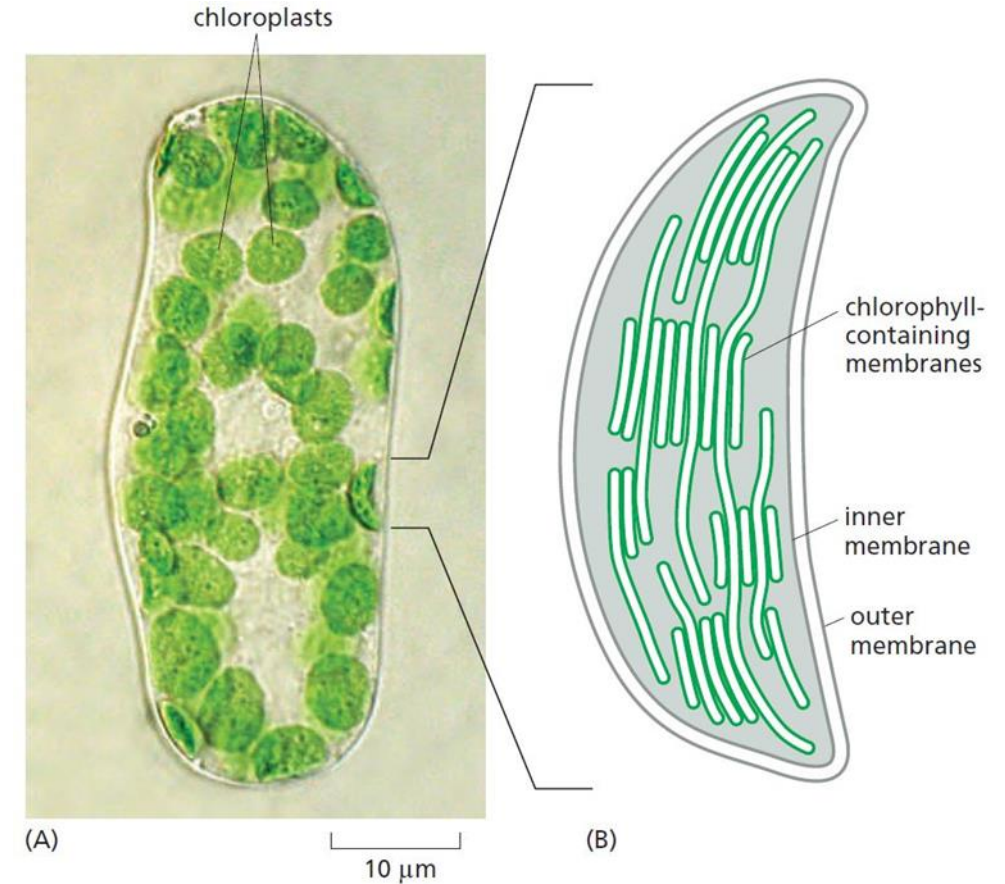


Figure 7. Chloroplasts surrounded by a double membrane (an outer and inner membrane).

1.3 Chloroplasts

- Chloroplast contains an extensive interconnected membrane bound sacs called **Thylakoids**.
- Thylakoids form stacks called **Grana** (singular Granum) and are embedded in a matrix called **Stroma**.
- Two Grana may be connected by a **Lamella**.
- The Thylakoid membranes contain green pigment (Chlorophyll) and other pigments that absorb light.

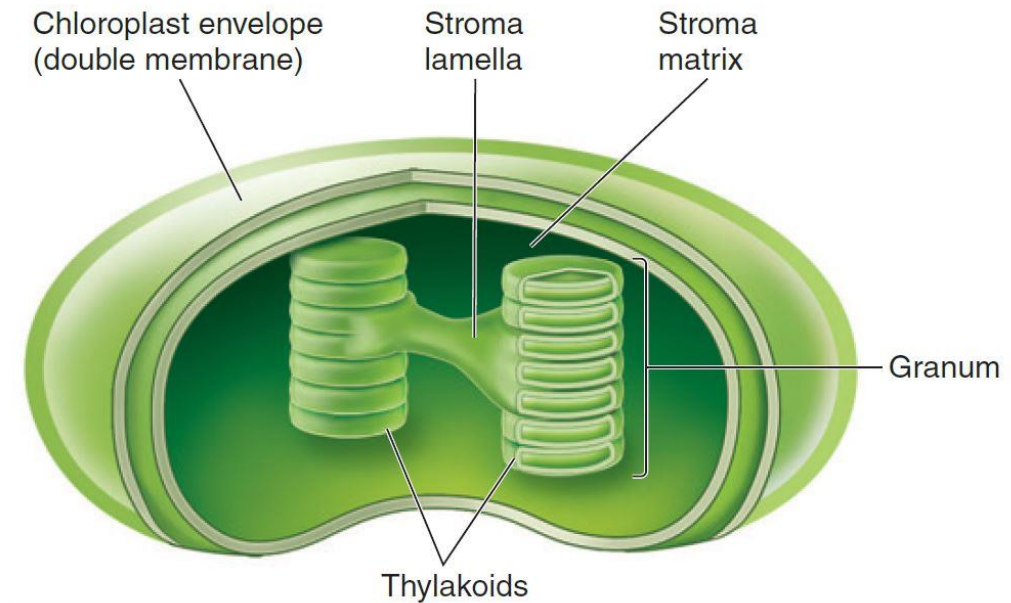


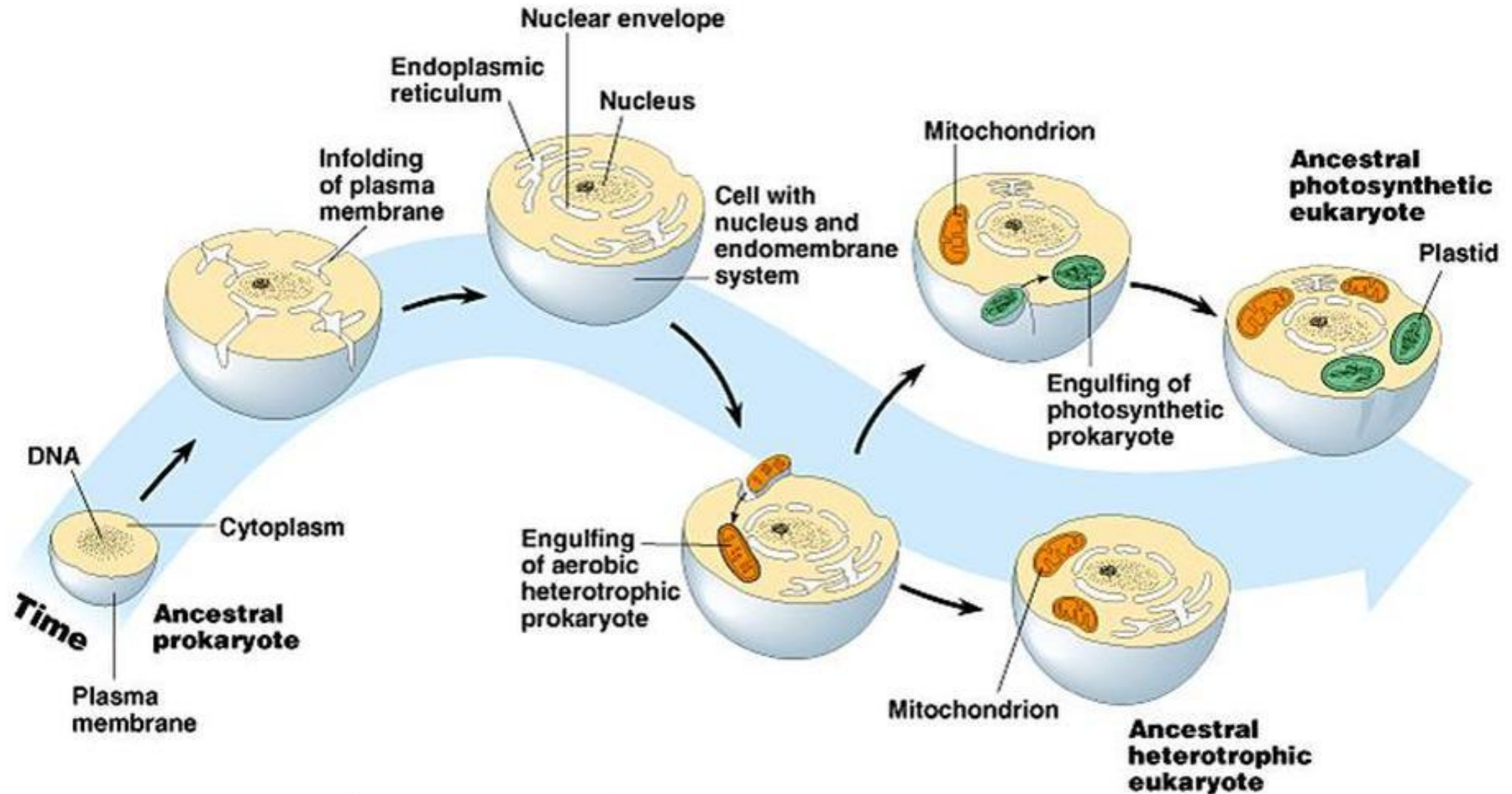
Figure 8. Chloroplast structure.

Cyanobacteria and Chloroplasts

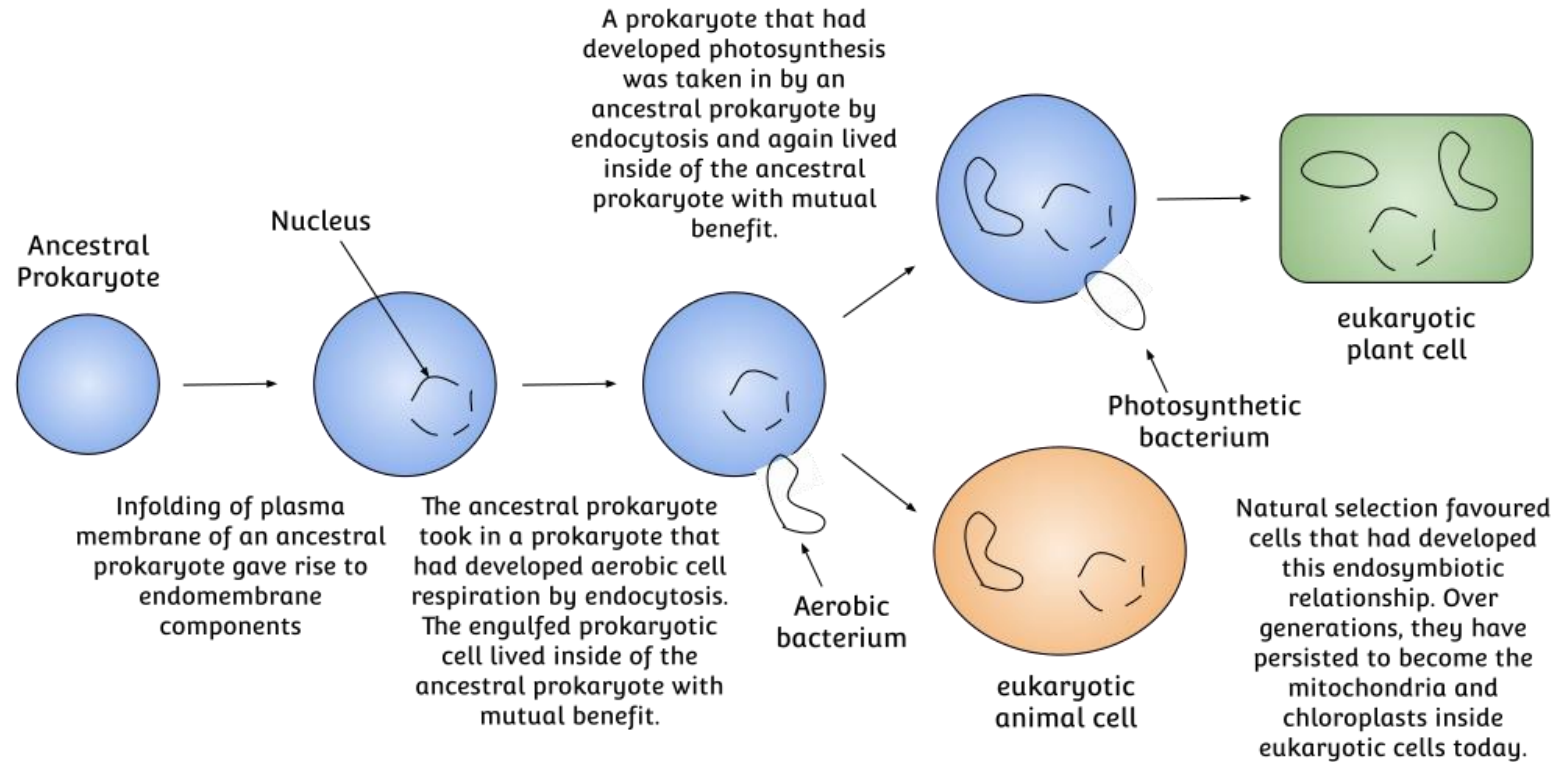


- ❑ Cyanobacteria are globally widespread photosynthetic prokaryotes and are major contributors to global biogeochemical cycles. They are the only oxygenic photosynthetic prokaryotes, and prosper in diverse and extreme habitats. They are among the oldest organisms on Earth with fossil records dating back 3.5 billion years. Since then, cyanobacteria have been essential players in the Earth's ecosystems.
- ❑ Cyanobacteria are the first organisms known to have produced oxygen.
- ❑ By producing and releasing oxygen as a byproduct of photosynthesis, cyanobacteria are thought to have converted the early oxygen-poor, reducing atmosphere into an oxidizing one, causing the Great Oxidation Event and the "rusting of the Earth", which dramatically changed the composition of the Earth's life forms and led to the near-extinction of anaerobic organisms.
- ❑ Cyanobacteria use the energy of sunlight to drive photosynthesis, a process where the energy of light is used to synthesize organic compounds from carbon dioxide.
- ❑ Chloroplasts have many similarities with cyanobacteria, including a circular chromosome, prokaryotic-type ribosomes, and similar proteins in the photosynthetic reaction center.
- ❑ The endosymbiotic theory suggests that photosynthetic bacteria were acquired (by endocytosis) by early eukaryotic cells to form the first plant cells.
- ❑ Therefore, chloroplasts may be photosynthetic bacteria that adapted to life inside plant cells.
- ❑ Like mitochondria, chloroplasts still possess their own DNA, separate from the nuclear DNA of their plant host cells and the genes in this chloroplast DNA resemble those in cyanobacteria.

Endosymbiosis (Mitochondria and Chloroplasts)



Endosymbiosis (Mitochondria and Chloroplasts)



Chloroplast and Mitochondrial Evidence

- They have double membranes
- They can only be produced by division of pre-existing mitochondria and chloroplasts
- They have their own DNA which is naked and circular
- They have ribosomes which are 70S in size

Organelles of the Eukaryotic Cells



- The Organelles are small, specialized structures in cells which perform specific functions.
- The cell organelles are of three types.

Those containing a, i) Double membrane, ii) Single membrane or iii) No membrane.

1. Double membrane bound cell organelles

1.1 Nucleus

1.2 Mitochondria

1.3 Chloroplast

2. Single membrane bound cell organelles

2.1 Endoplasmic reticulum

2.2 Golgi apparatus

2.3 Vacuole

2.4 Lysosomes and endosomes

2.5 Peroxisomes

3. Cell organelles with out the membrane

3.1 Ribosomes

3.2 Centrosomes (Centriole)

3.3 Cytoskeleton (Microtubules, actin filaments and intermediate filaments)

Single membrane Cell Organelles



2.1 Endoplasmic Reticulum (ER)

- Endoplasmic Reticulum (ER) is a network of membrane system constituting the channels and is separated from cytoplasm by a single membrane.
- The extensive network of closed, flattened membrane-bound sac like structures of ER are called **Cisternae**, whose interior is called **Cisternal space** or **Lumen**.
- ER is involved in Protein synthesis, Lipid metabolism and Calcium storage.

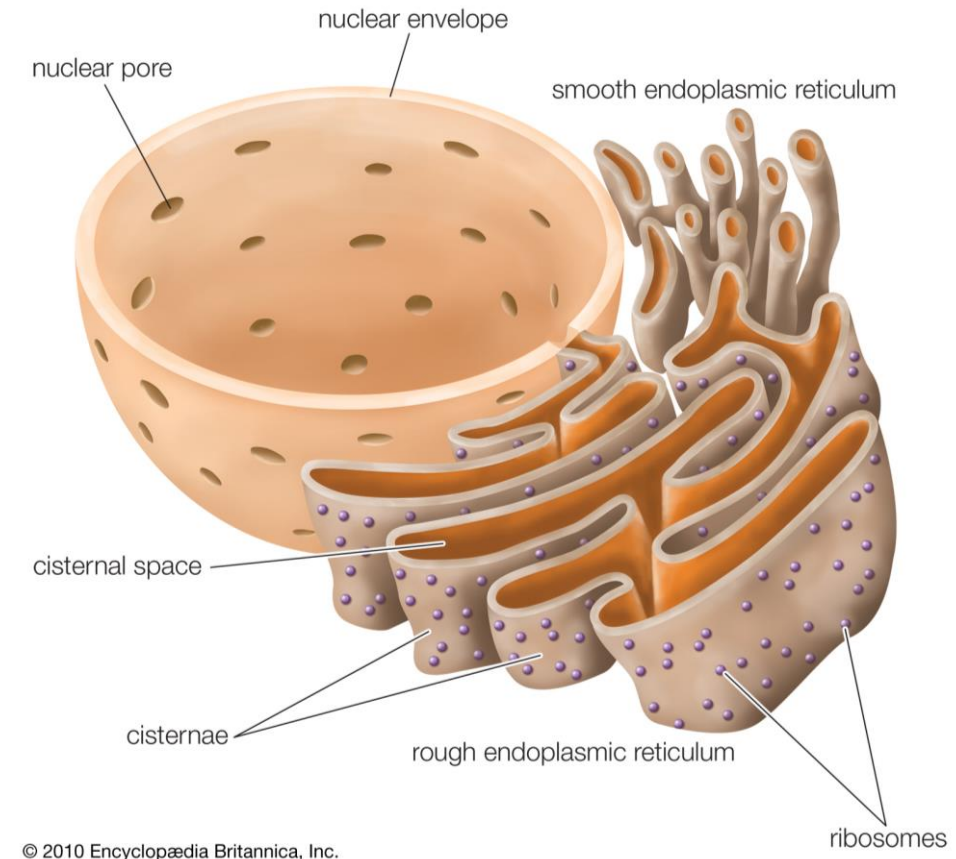
There are two types of Endoplasmic Reticulum

1. Smooth Endoplasmic Reticulum (SER).

The ER without the Ribosomes are called Smooth ER. The synthesis of fatty acids and phospholipids takes place in the smooth ER.

2. Rough Endoplasmic Reticulum (RER).

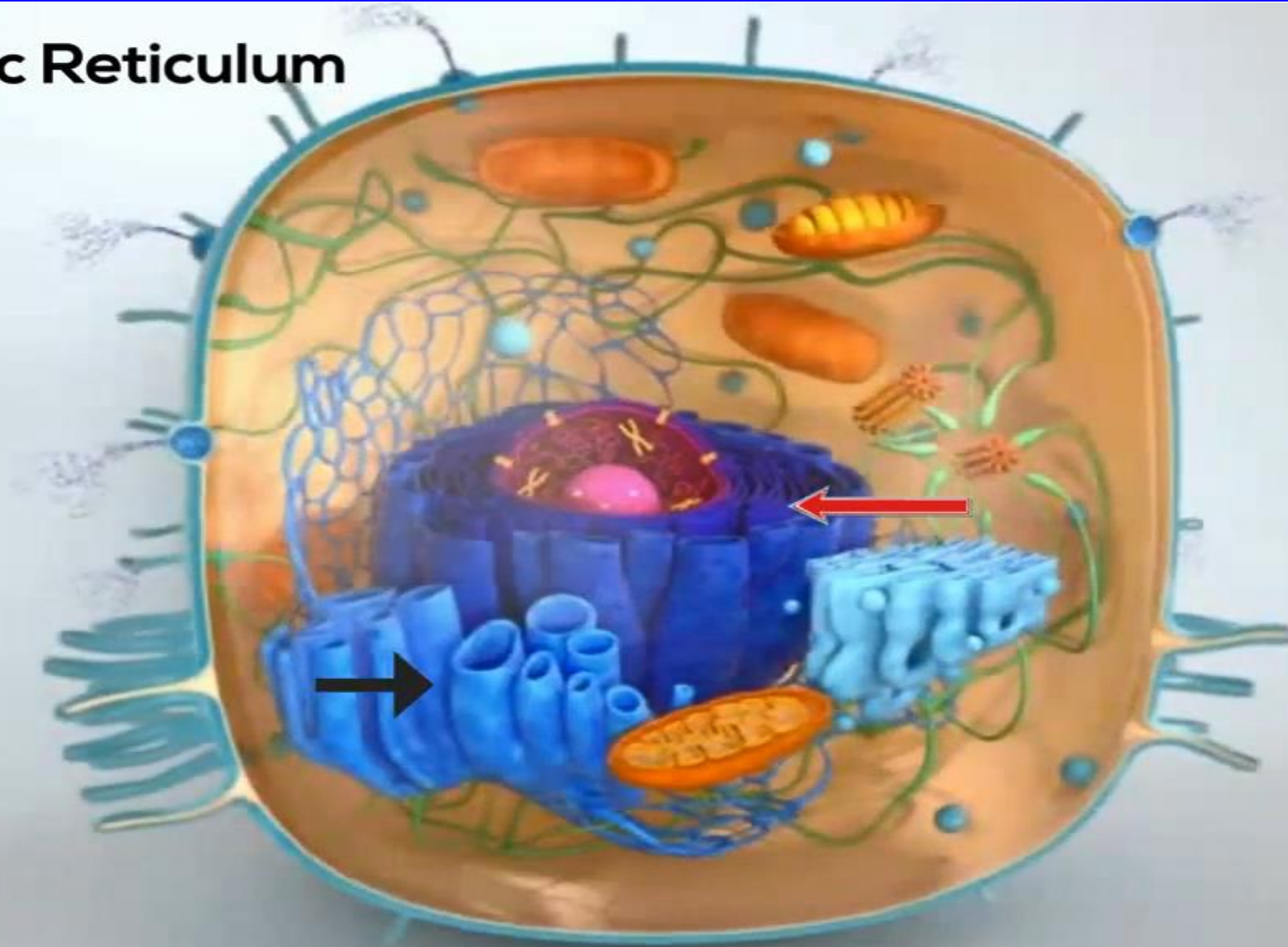
The ER attached with Ribosomes are called Rough ER. They synthesize membrane and organelle proteins and virtually all proteins required by the cell. They are involved in the synthesis, folding, modification and transport of proteins to rest of the cell.



**Fig. 1. Endoplasmic Reticulum (ER).
Smooth and Rough ER**

2.1 Endoplasmic Reticulum

Endoplasmic Reticulum



Important Links to see Videos



❑ **ER:**

❑ <https://www.youtube.com/watch?v=an7tpWR16mo>

2.2 Golgi Apparatus

Golgi apparatus, is named after its discoverer, Camillo Golgi (1906 Nobel prize winner).

- The Golgi apparatus (also called Golgi complex or Golgi body) is a membrane-bound organelle.
- It is made up of a series of flattened stacked pouches called **Cisternae**. It is located in the Cytoplasm next to the Endoplasmic Reticulum and near the Nucleus
- It helps in the processing and packaging of proteins and lipid molecules.
- Proteins after synthesis in Rough ER leaves the organelle within a small membrane bound **Vesicles** (which originate from Rough ER).
- These **Vesicles** act as transport vesicles which carry the proteins to **Golgi complex**.
- Stack of Golgi Cisternae has three defined regions
i) cis (face towards ER), ii) the medial, iii) the trans.

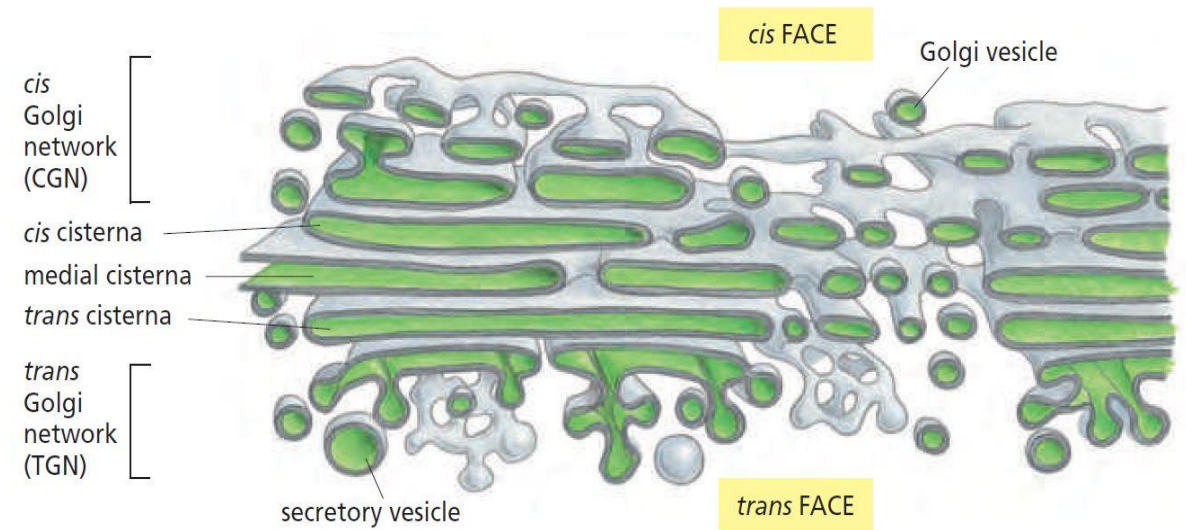
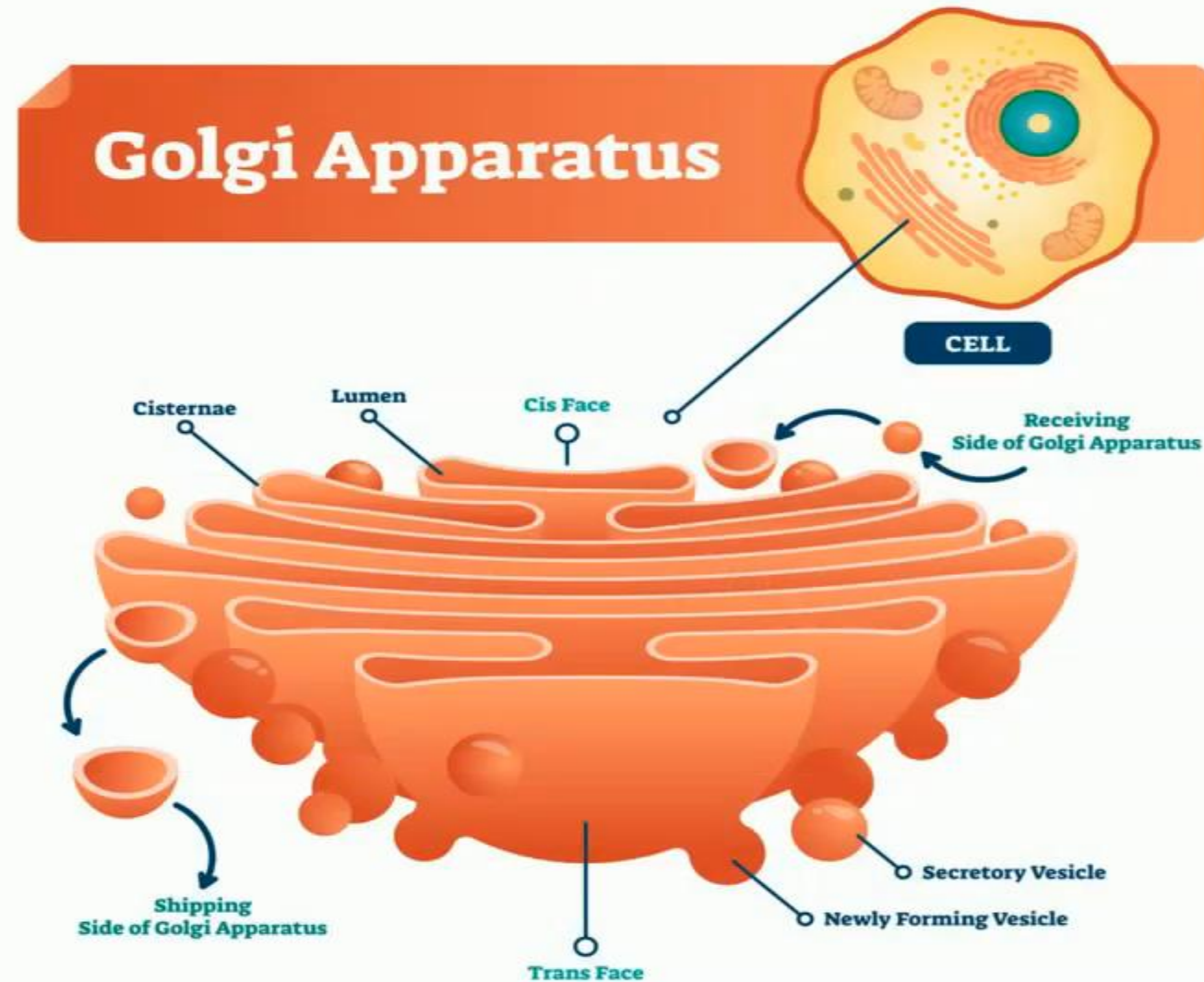


Fig. 3. The Golgi Apparatus.

2.2 Golgi Apparatus



Important Links to see Videos



- ❑ **Golgi Apparatus:**
- ❑ <https://www.youtube.com/watch?v=d0QljHf7REM>

2.3 Vacuole (Structure)

- **Vacuoles are membrane-bound organelles found in both animal and plant cells.**
- **The vacuoles in plant cells are larger than those in the animal cells. The plant vacuole can occupy more than 80% of the volume of the cell.**

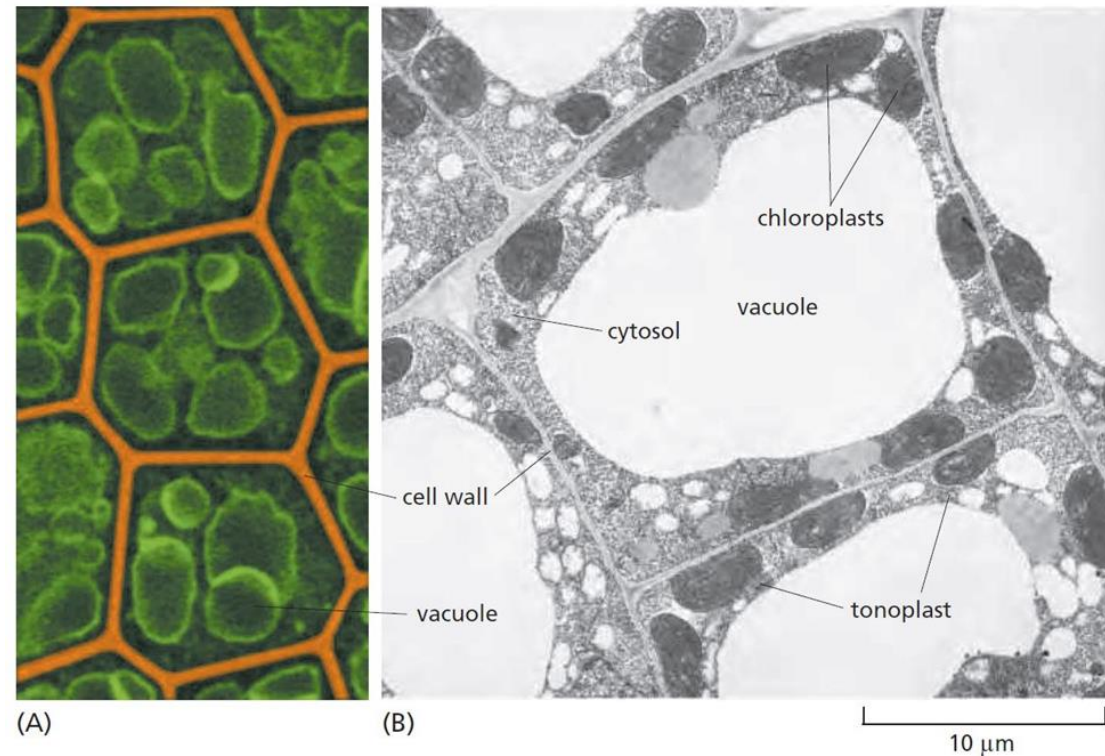


Fig. 4. Plant Cell Vacuole.

A) Confocal Microscopic image of Arabidopsis (Plant, mustard family) cells showing Aquaporin (a yellow fluorescent protein) in Vacuole membrane (green). The cell walls are shown in orange colour. Each cell contains several large vacuoles.

B) Electron micrograph of Tobacco Leaf cells showing an Enormous vacuole pressed against the cell wall.

2.3 Vacuole (Function)

The important functions of Vacuole are

1. **Storage:** A vacuole stores salts, minerals, pigments and proteins within the cell.
2. **Turgor Pressure:** The vacuoles are completely filled with water and exert force on the cell wall. This is known as **Turgor Pressure**. It provides shape to the cell and helps to withstand the extreme conditions.
3. **Endocytosis and Exocytosis:** The substances are taken in by a vacuole through **Endocytosis** and excreted through **Exocytosis**.
4. **Lumen** (the 3-D interior space) of a vacuole in plant cells contains degradative enzymes. Therefore, plant vacuoles also have a degradative function, similar to that of Lysosomes in animal cells.

2.4 Lysosomes and endosomes



- Lysosomes are membrane-bound cell organelles, which contain digestive enzymes due to which they are also called as **Cell Stomachs**.
- The major function of Lysosomes is digestion and removal of waste.
- Lysosomes are plentiful in cells (**such as macrophages**) involved in digesting and destroying other cells.
- Lysosomes contain enzymes such as
 - i) **Nucleases** which degrade RNA and DNA into their respective mononucleotides.
 - ii) **Proteases** which degrade a variety of proteins and peptides.
- Isolating such potentially destructive enzymes in the Lysosome, protects the cytosol.

2.4 Lysosomes and Endosomes

Primary Lysosomes – They are small vesicles produced by Golgi apparatus.

Secondary Lysosomes – They are formed when Primary Lysosomes fuse with **Endosomes**.

Lysosomes digest the material by 3 pathways: Endocytosis, Phagocytosis and Autophagy

- 1. Endocytosis** is the process of actively transporting molecules **into the cell** by engulfing it with its membrane.
- 2. Phagocytosis** is a specific form of endocytosis, by which the cells internalise external solid matter, including microbial pathogens.
- 3. Autophagy** (Self eating) is a self-digesting mechanism responsible for removal of damaged organelles or other cellular components by lysosome.

Endosomes are membrane-bound vesicles involved in

- i. Sorting and delivering of the internalized material (such as nutrient uptake) from the cell surface and
- ii. the transport of material from Golgi complex to Lysosome or Vacuole.

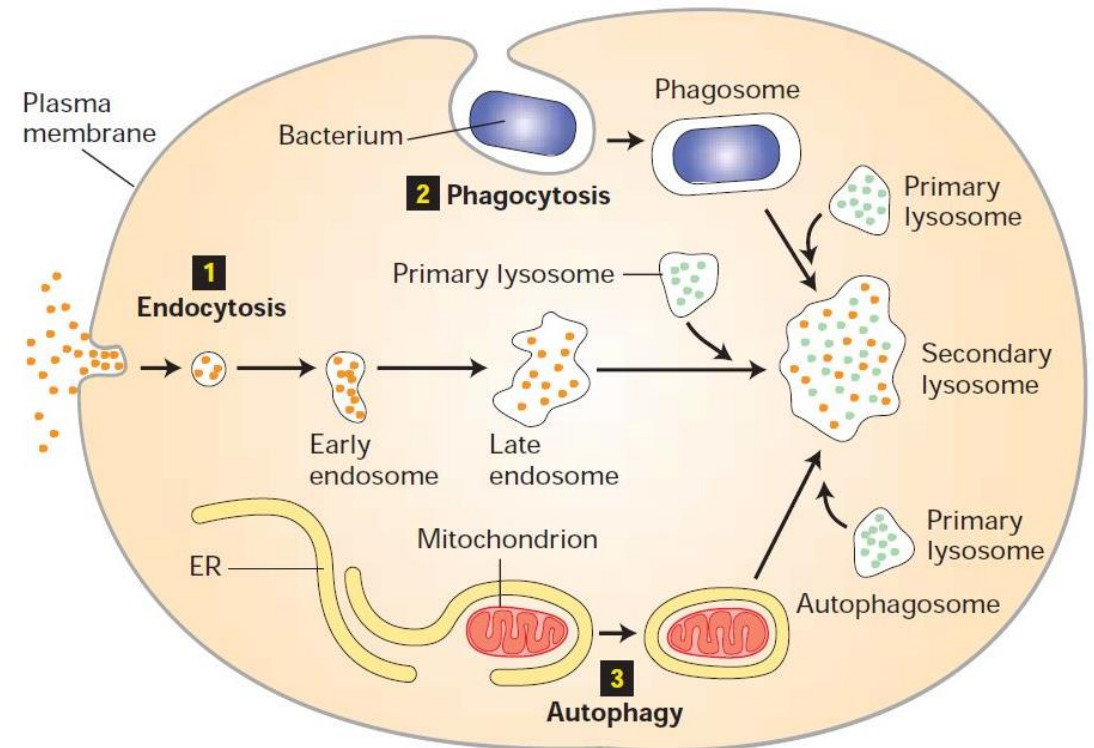
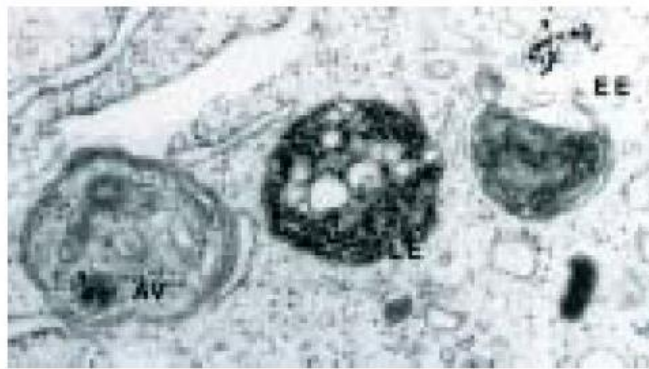


Fig. 5. Schematic view of Three pathways by which materials are moved to Lysosomes.

**Credit: Molecular Cell Biology-Lodish*

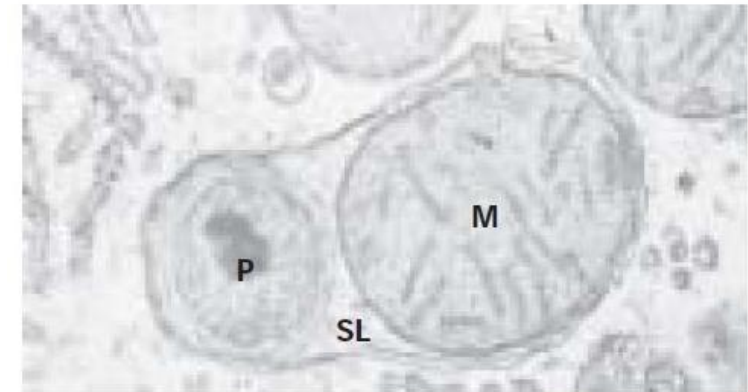
2.4 Lysosomes and Endosomes



0.1 μm

Fig. 6A. Electron micrograph of a section of a **mammalian cell** with internalized gold particles coated with the egg protein, Ovalbumin.

Gold-labeled Ovalbumin (black spots) is found in **Early Endosomes (EE)** and **Late Endosomes (LE)**, but very little is present in **Autophagosomes (AV)**.

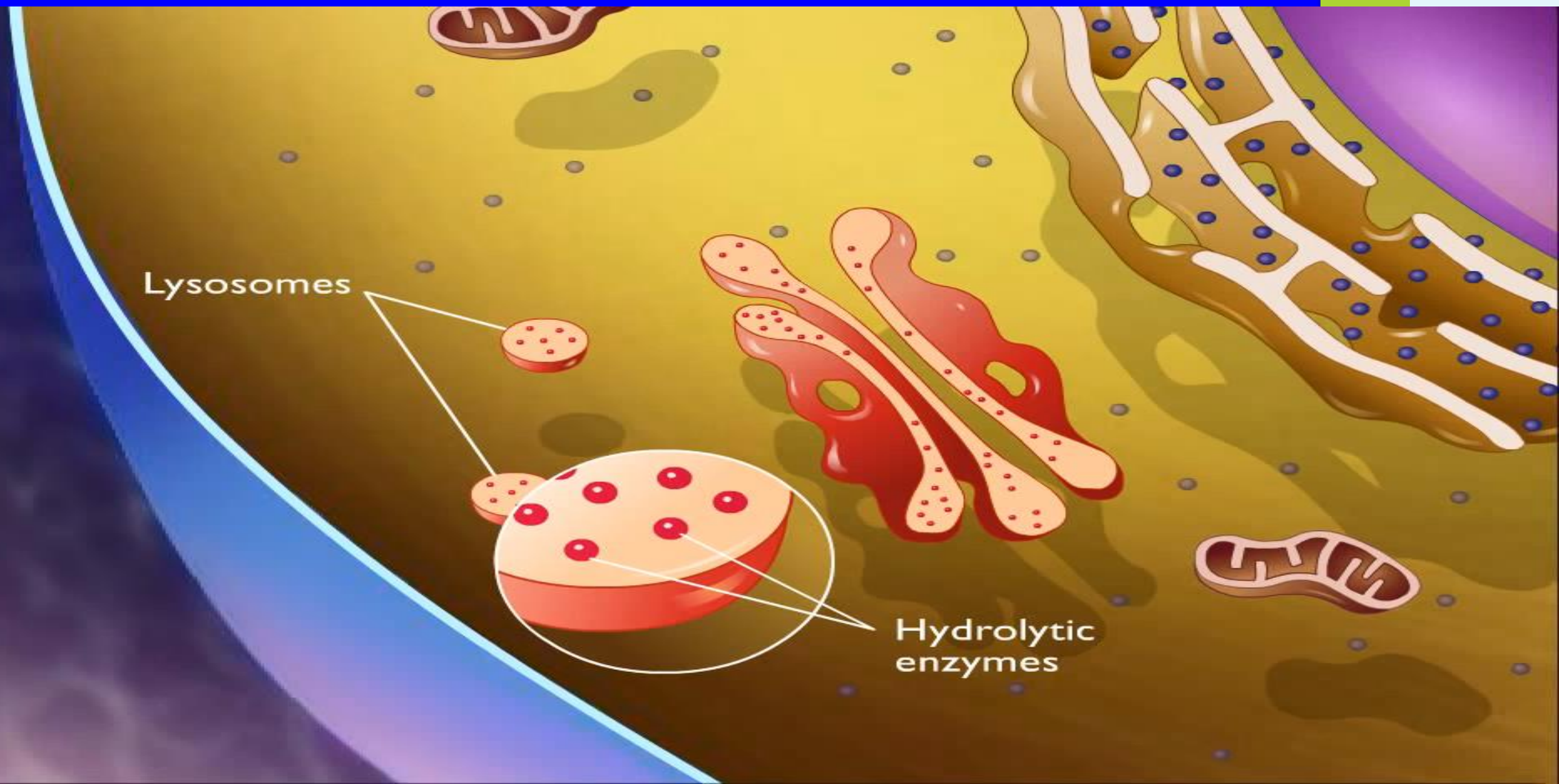


1 μm

Fig. 6B. Electron micrograph of a section of **Rat Liver cell** showing a **Secondary Lysosome** containing a **Mitochondrion (M)** and a **Peroxisome (P)**.

**Credit: Molecular Cell Biology-Lodish*

Lysosomes



Important Links to see Videos



- ❑ **Lysosomes:**
- ❑ https://www.youtube.com/watch?v=cA-Ou_t2sag

2.5 Peroxisomes (Structure)



- Peroxisomes are small, **single membrane-enclosed organelles** which contain enzymes involved in a variety of metabolic reactions.
- All animal cells (except Erythrocytes, the Red Blood cells) and many plant cells contain peroxisomes, a class of roughly spherical organelles, 0.2–1.0 μm in diameter.



- *Credit: Molecular Biology of the Cell - Alberts*

2.5 Peroxisomes (Functions)

- Peroxisomes contain several **Oxidase** enzymes, which use molecular oxygen (O_2) to oxidize organic substances (designated as R here), leading to the formation of **Hydrogen peroxide** (H_2O_2).



- When excess H_2O_2 accumulates in the cell, another enzyme, **Catalase** converts it into H_2O by the reaction.



- In humans, peroxisomes are primarily associated with **Lipid metabolism**, While in photosynthetic plant tissues, their most prominent role is in **Photorespiration**.
- Peroxisomes have the ability to adapt to the changing conditions.
e.g. Yeasts grown on sugar, have few small peroxisomes, but when they are grown on methanol, numerous large peroxisomes are formed that oxidize methanol.

Cell organelles without membrane



3.1 Ribosomes

- A ribosome is a complex molecular machine found inside the living cells that produces proteins from amino acids during the process called **Translation or Protein Synthesis**.
- Ribosomes are found in both Prokaryotic and Eukaryotic cells.
- Ribosomes typically contain about 80% of the total cellular RNA. They are complex structures, each consisting of more than 50 different proteins and RNA molecules (Table 1).
- A functional ribosome consists of two subunits, a Large Ribosomal Subunit and a Small Ribosomal Subunit.
- The size of Ribosomes and their RNA components are given in **Svedberg (S)** Units (a measure of how rapidly it sediments in a centrifugal field).

Table 1. Composition of ribosomes in Bacterial and Eukaryotic cells.

Cell Type	Ribosome Size	Subunit	rRNA Component	Proteins
Bacterial	70S	Large (50S)	23S (2900 nucleotides), 5S (120 nucleotides)	31
		Small (30S)	16S (1500 nucleotides)	21
Eukaryotic	80S	Large (60S)	28S (4700 nucleotides), 5.8S (160 nucleotides), 5S (120 nucleotides)	49
		Small (40S)	18S (1900 nucleotides)	33

It is important to note that Svedberg units are not additive.

3.1 Structure of Ribosome

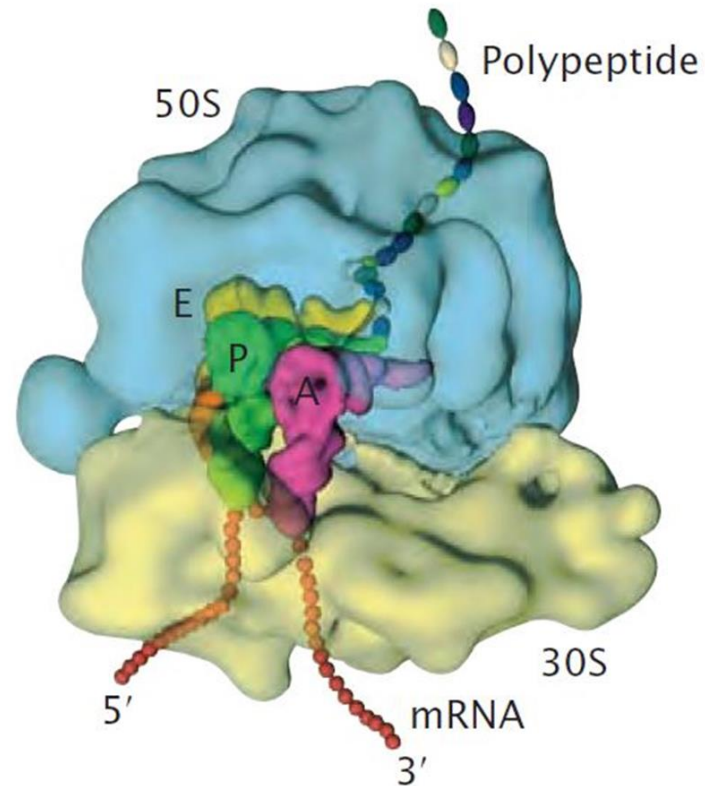


Fig. 2. Structure of the Bacterial Ribosome. Low-resolution model of Ribosome, showing the A (Amino acid binding), P (peptide and also tRNA binding site) and E (the exit site) where the growing polypeptide chain reside.

3.2 The Centrosome (Structure)

Centrosomes are made up of **two, Barrel-shaped clusters of Microtubules** called “**Centrioles**” and a complex of proteins which help in the formation of additional microtubules.

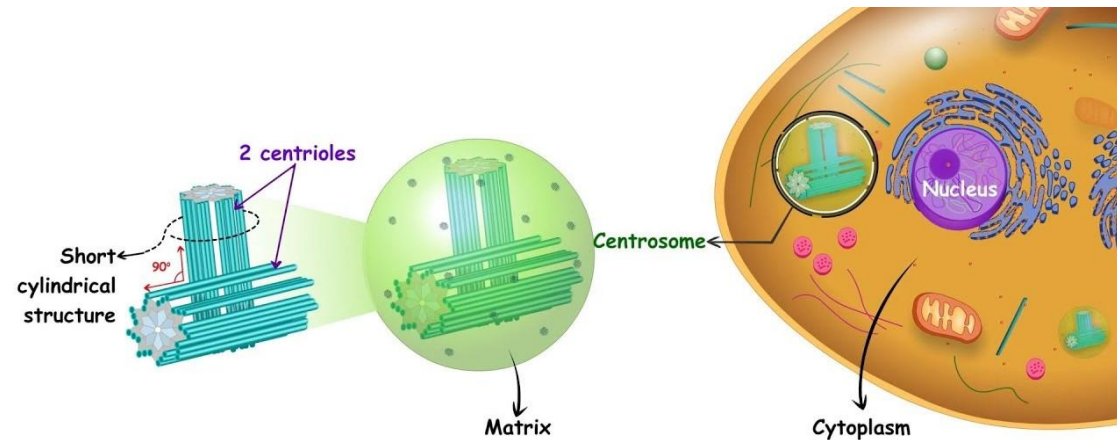


Fig. 3. Centrosome

Centrosomes are not found in higher plants, but it can be found in some lower plants.
e.g. Mosses and Ferns.



Fig. 4. a) Moss



b) Fern

3.2 The Centrosome (Function)



- Centrosomes play an important role during **cell division** as the organizers of the microtubules that pull sister chromatids apart, ensuring that each daughter cell gets a full compliment of the parent cell's DNA.
- Apart from cell division, they are also involved in the formation of **Cilia and Flagella** and thus contribute to the cell movement.
- Centrosomes are organelles that serve as **Microtubule-Organizing Centers (MTOC)**.

3.3 The Cytoskeleton: Components, Structure & Functions

- The cytoskeleton is a network of filaments and tubules that extends throughout the cell, through the Cytoplasm.
- It is found in all cells, though the proteins that it is made up of, vary among the organisms.
- Three types of filaments compose the Cytoskeleton:
 1. **Actin filaments**, also called microfilaments, provides mechanical support, determines cell shape and allows the movement of the cell surface.
 2. **Intermediate filaments** (IFs), their function is purely structural. They bear tension, thus maintaining the shape of the cell, and anchor the nucleus and other organelles in place.
 3. **Microtubules**, play important roles in cell movement, cell division and cellular transport.

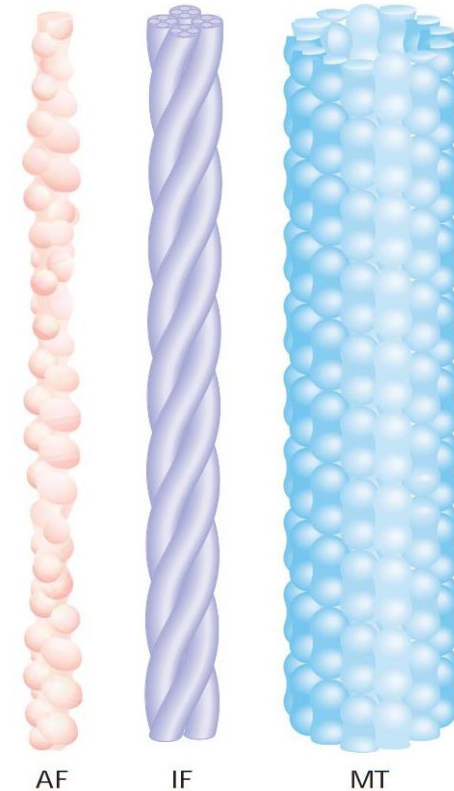


Fig. 5. Three types of filaments forming Cytoskeleton.
Actin filament (AF),
Intermediate filament (IF),
Microtubule (MT).

3.3 The Cytoskeleton: Components and Structure

- 1. Microfilaments:** They are 8-9 nm in diameter and have a twisted two-stranded structure. They are composed of the Actin protein.
- 2. Intermediate filaments:** are about 8-12 nm wide. They are called intermediate, because their size is in-between the sizes of microfilaments and microtubules. They are made up of different proteins, such as Keratin Vimentin, Desmin and Lamin.
- 3. Microtubules:** are **Hollow tube-like structures**, 24 nm in diameter. They are made up of **Tubulin** protein.

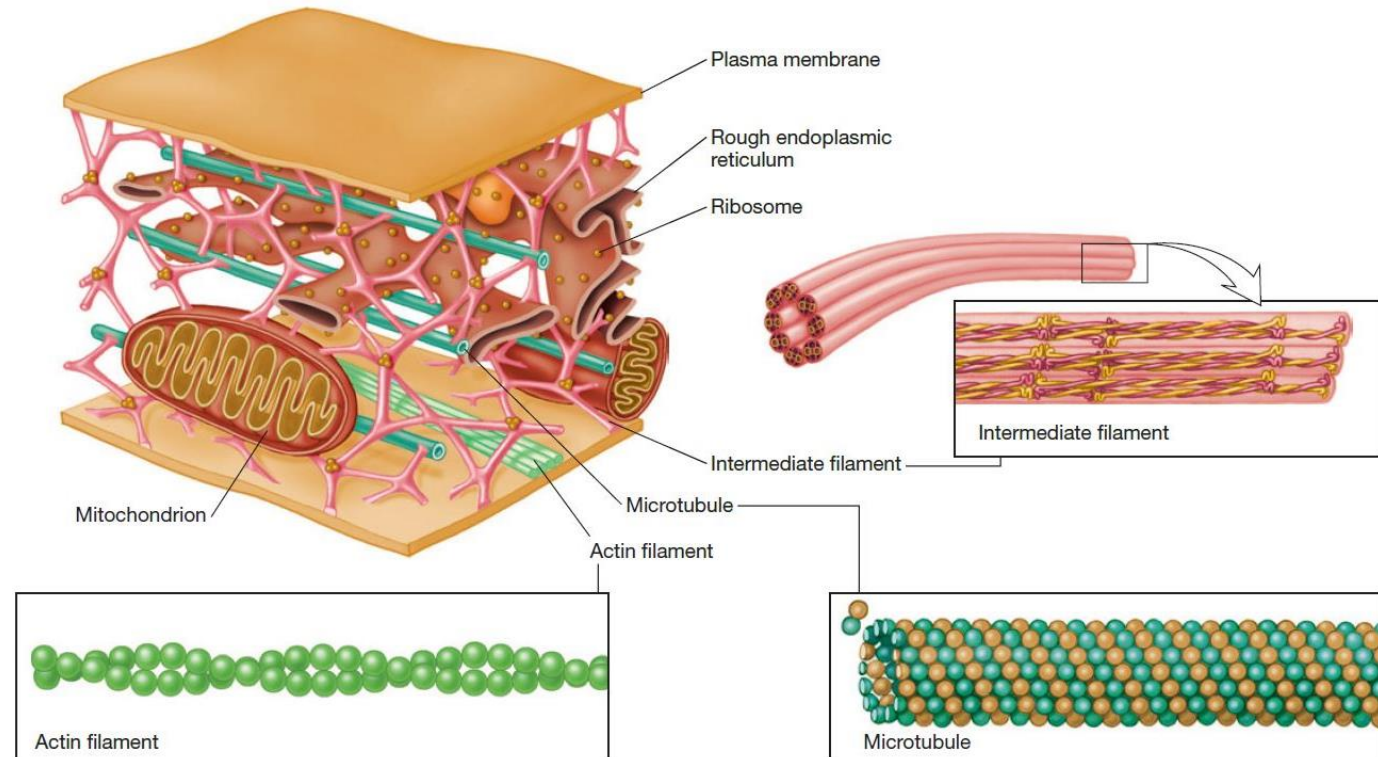
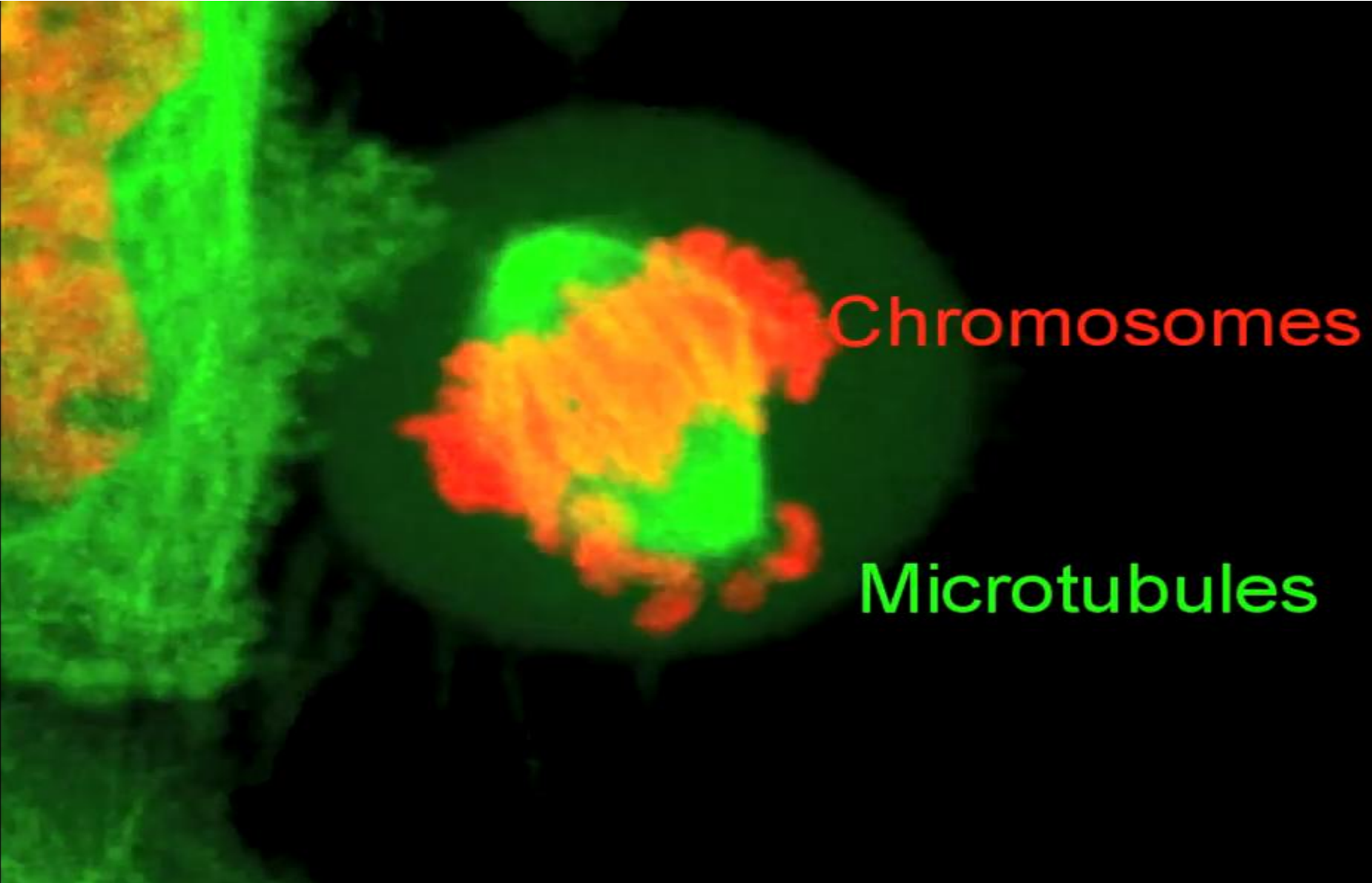


Fig. 6. The Eukaryotic Cytoskeleton.

Cytoskeleton



Important Links to see Videos



- ❑ **Cytoskeleton:**
- ❑ <https://www.youtube.com/watch?v=YTv9ItGd050>

Various organelles in a human cell and its major function(s)



- 1. Cell membrane** – (is a membrane that separates the interior of all cells from the outside environment (the extracellular space) and protects the cell from its environment; acts as a barrier for cell contents. It consists of double phospholipid layer and monolayer of protein scattered around phospholipid layer. The other components in plasma membrane are cholesterol and glycoproteins).
- 2. Cytosol** – (fluid that contains organelles; represents the material outside the nucleus and inside the cell membrane)
- 3. Cytoskeleton** – (a network of protein structures that extend throughout the cytoplasm; provides the cell with an internal framework (e.g. microfilaments and microtubules))
- 4. Nucleus** – (control center of the cell; contains genetic material; It is the center of the cell because it contains genetic material (DNA). It consists of three main regions: the nuclear membrane, the nucleolus and chromatin.
Nuclear membrane: Nuclear membrane serves as a barrier of nucleus. It consists of a double phospholipid membrane and contains nuclear pores that allow for the exchange of material with the rest of the cell.
Chromatin: It is composed of DNA and protein scattered throughout the nucleus. Chromatin condenses to form chromosomes when the cell divides)
- 5. Nucleolus** – (functions as site of ribosome production; ribosomes then migrate to the cytoplasm through nuclear pores)
- 6. Ribosome** (little dots) – sites of protein synthesis in the cell.

Various organelles in a human cell and its major function(s)



7. **Rough endoplasmic reticulum** – (carry ribosomes that represent sites of protein synthesis)
8. **Smooth endoplasmic reticulum** – (function in cholesterol synthesis and breakdown, fat metabolism, and detoxification of drugs)
9. **Golgi apparatus** (or "Golgi body") – (modifies and packages proteins, secrete vesicles, plasma membrane components and lysosomes)
10. **Mitochondrion** – (powerhouse of the cell; generate ATP for cellular activities)
11. **Lysosomes** – (suicidal bag of the cell; contain enzymes that digest non-usable materials within the cell)
12. **Peroxisomes** – (detoxify harmful substances and break down free radicals; involved in catabolism of very long chain fatty acids,)
13. **Centrioles** – (involved in the organization of mitotic spindle fibers during cell division and in the completion of cytokinesis)
14. **Centrosomes** – (serves as the main microtubule organizing center (MTOC) of the animal cell, as well as a regulator of cell-cycle progression; are composed of two centrioles arranged at right-angles to each other)
15. **Vesicles** – (involved in metabolism, transport, buoyancy control, and temporary storage of food and enzymes)

Important Links to see Videos



☐ Cells:

- ☐ <https://www.youtube.com/watch?v=URUJD5NEXC8>
- ☐ <https://www.youtube.com/watch?v=FYwB1EETakY>
- ☐ <https://www.youtube.com/watch?v=zZtcMBTQaS4>
- ☐ <https://www.youtube.com/watch?v=Pxujitlv8wc>

☐ DNA to protein:

- ☐ <https://www.youtube.com/watch?v=gG7uCskUOrA>